



**Kandivli Education Society's  
B. K. SHROFF COLLEGE OF ARTS &  
M. H. SHROFF COLLEGE OF COMMERCE**

**An Autonomous College**

**NAAC Re-accredited 'A' Grade**

ISO 9001 : 2015 Certified • 'Best College 2017-18' award from University of Mumbai

**PROJECT REPORT**

**ON**

**IOT-BASED SMART HOME SECURITY SYSTEM**

**IN THE PROGRAMME**

**BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)**

**SUBMITTED BY**

**DHARMI BHAYANI**

**B.Sc. (IT)**

**TDIT006A**

**SEMESTER VI**

**UNDER THE GUIDANCE OF**

**DR. VISHESH SHRIVASTAVA**

**ACADEMIC YEAR**

**2025– 2026**



**Kandivli Education Society's  
B. K. SHROFF COLLEGE OF ARTS &  
M. H. SHROFF COLLEGE OF COMMERCE**

**An Autonomous College**

**NAAC Re-accredited 'A' Grade**

ISO 9001 : 2015 Certified • 'Best College 2017-18' award from University of Mumbai

**CERTIFICATE**

This is to certify that **Mrs.DHARMI BHAYANI** of **THIRD** year of **Bachelor of Science in Information Technology**, Div:A, Roll No.**TDIT006A** of Semester **VI** (2025 - 2026) has successfully completed the Project on the topic **IOT-BASED SMART HOME SECURITY SYSTEM** as per the guidelines of KES' Shroff College of Arts and Commerce, Kandivali (W), Mumbai-400067.

**Teacher In-charge**

**Dr. Vishesh Shrivastava**

**Principal**

**Dr. Lily Bhushan**

## PROFORMA FOR THE APPROVAL PROJECT PROPOSAL

**(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)**

PRN No.: .....

Roll no: TDIT006A

1. Name of the Student

DHARMI NARESHBHAI BHAYANI

---

2. Title of the Project

IOT-BASED SMART HOME SECURITY SYSTEM

---

3. Name of the Guide

DR. VISHESH SHRIVASTAVA

---

Signature of the Student

Signature of the Guide

Date: .....

Date: .....

Signature of the

Coordinator

Date:

.....

## ABSTRACT

The Internet of Things (IoT) based Smart Home Security System is an advanced technology that allows homes to be monitored and protected using internet-connected devices. Smart security systems use sensors, cameras, and microcontrollers to detect unusual activities and provide real-time alerts to homeowners. Motion detection and intrusion monitoring are important features of modern smart home security systems, which help in identifying unauthorized access and ensuring the safety of the home environment.

This project focuses on developing an IoT-based smart home security system that can monitor different areas of a house and provide automatic alerts when suspicious activities are detected. The system uses sensors such as motion sensors, door sensors, and cameras connected to a microcontroller and internet network. These devices continuously collect data and send notifications to the user through a mobile application or web interface. By using IoT technology, the system can provide real-time monitoring and quick response to potential security threats.

IoT-based smart home security systems have wide applications in residential buildings, apartments, and smart homes. They help homeowners monitor their property remotely and improve safety and convenience. The system can detect motion, control security devices, and notify users instantly when any abnormal event occurs. This project demonstrates how IoT technology can be applied to improve home security and automate monitoring processes.

The rapid development of IoT technology has greatly improved the ability of smart devices to communicate and share information through the internet. Smart home security systems use connected sensors and devices to collect and analyze data from the home environment. In this project, an IoT-based system is designed to detect intrusion, monitor surroundings, and provide alerts to users in real time.

The system can automatically detect multiple security events such as motion detection, door opening, or unusual activities inside or outside the home. By combining IoT technology with smart sensors and automation, this project aims to enhance home security and support modern smart living. The system provides a reliable and efficient way to protect homes, improve safety, and enable remote monitoring through connected devices.

## **ACKNOWLEDGEMENT**

I would like to express my heartfelt gratitude to all those who supported me throughout the completion of this project titled “IOT-Based Smart Home Security System.”

First and foremost, I am deeply thankful to my project guide Dr Vishesh Shrivastava, whose valuable guidance, constant encouragement, and expert advice have been a great source of inspiration during this work.

I am also thankful to KES Shroff College for providing us with the necessary resources and facilities to carry out our research and implementation effectively. Furthermore, I would like to acknowledge the contributions of our friends and family for their unwavering support and understanding during this endeavor.

Finally, I express my deepest gratitude to my Senior for their motivation, and moral support, which made this work possible.

Thank you.

Dharmi Bhayani

## **DECLARATION**

I hereby declare that the project entitled, “IOT-Based Smart Home Security System” done at KES Shroff College, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

I further declare that this project is the result of my own independent work and research. All the information, data, and results presented in this report are true to the best of my knowledge and belief.

Dharmi Bhayani

## Table of Contents

|                                                                |              |
|----------------------------------------------------------------|--------------|
| <b>Chapter 1: Introduction.....</b>                            | <b>09</b>    |
| 1.1 Significance of IOT-Based Smart Home Security System ..... | 10           |
| 1.2 Objectives of this Learning Management System.....         | 11           |
| 1.3 Scope of "IOT-Based Smart Home Security System" .....      | 12           |
| <b>Chapter 2: System Analysis.....</b>                         | <b>13</b>    |
| 2.1 Existing System: .....                                     | 13           |
| 2.2 Proposed System: .....                                     | 14-15        |
| 2.3 Requirement Analysis: .....                                | 16-17        |
| 2.4 Hardware Requirements .....                                | 18           |
| 2.5 Software Requirements.....                                 | 19           |
| 2.6 Justification of Platform. ....                            | 20-21        |
| <b>Chapter 3: System Design.....</b>                           | <b>22-26</b> |
| 3.1 Module Division .....                                      | 27-39        |
| 3.2 Data Dictionary .....                                      | 30-34        |
| 3.3 ER Diagrams .....                                          | 35           |
| 3.4 Data Flow Diagrams.....                                    | 36           |
| 3.4.1 System Flow diagram.....                                 | 36           |
| 3.4.2 Class diagram .....                                      | 37           |
| 3.4.3 Sequence diagram.....                                    | 38           |
| 3.4.4 State Chart diagram....                                  | 39           |
| 3.4.5 Use-Case Diagram.....                                    | 40           |
| <b>Chapter 4: Implementation and Testing.</b>                  |              |
| 4.1 Code.....                                                  | 41           |
| 4.2 Testing Approach .....                                     | 47-48        |

## **Chapter 5: Results and Discussions**

|                                  |       |
|----------------------------------|-------|
| Functionality Evaluation: .....  | 49-50 |
| User Experience Assessment:..... | 51-52 |
| Data Security and Integrity..... | 53    |

## **Chapter 6: Conclusion and Future Work**

|                   |       |
|-------------------|-------|
| Conclusion.....   | 54-55 |
| Future Scope..... | 56-58 |
| Limitations.....  | 59-60 |

## **Chapter 7: References.....61**



## **Chapter 1: Introduction**

The Internet of Things (IoT) has become one of the most important technologies in the modern digital era, allowing everyday devices to connect, communicate, and share data through the internet. One of the most useful applications of IoT is in home automation and security systems, which help homeowners monitor and protect their houses more efficiently. When IoT technology is applied to home security, it enables smart devices such as sensors, cameras, and alarms to work together to detect suspicious activities and provide real-time alerts to users.

In an IoT-based smart home security system, multiple sensors and devices are connected to a central controller and communicate through the internet. These devices can monitor activities such as motion detection, door opening, or unusual movements within or around the home environment. When the system detects any unauthorized access or abnormal event, it can automatically send notifications or alerts to the homeowner through a mobile application or web interface. This allows users to monitor and control their home security system remotely from anywhere.

This project focuses on designing and implementing an IoT-based smart home security system that can automatically detect potential security threats and improve home safety. The system uses IoT devices such as motion sensors, cameras, and microcontrollers to collect and transmit data in real time. These devices continuously monitor the home environment and respond quickly when suspicious activities are detected. By using IoT technology, the system becomes more efficient, reliable, and capable of providing instant alerts and remote monitoring.

The main purpose of this project is to demonstrate how IoT technology can be used to build an intelligent home security system that improves safety and convenience for homeowners. Such systems have many real-world applications in residential buildings, apartments, offices, and smart homes. They help reduce security risks by providing continuous monitoring and quick responses to potential threats.

Through this project, we aim to explore how IoT devices and smart sensors can work together to create an automated security solution. By combining IoT technology with modern communication networks, the system provides a reliable and efficient way to protect homes and ensure safety. This project highlights the role of IoT in transforming traditional home security systems into smart, connected, and intelligent security solutions.

## 1.1 Significance of IOT-Based Smart Home Security System

The significance of an IoT-based Smart Home Security System lies in its ability to provide intelligent, connected, and real-time protection for modern homes. Traditional security systems mainly relied on basic alarms or manual monitoring, which often required constant human attention and could not provide immediate updates. With the integration of the Internet of Things (IoT), home security systems have become more advanced and efficient. IoT technology allows multiple smart devices such as sensors, cameras, alarms, and locks to communicate with each other through the internet, enabling homeowners to monitor and control their security systems remotely.

An IoT-based smart home security system plays an important role in detecting and preventing unauthorized access or suspicious activities within a home environment. By using devices such as motion sensors, door sensors, and surveillance cameras, the system can continuously monitor different areas of the house. When unusual activity is detected, the system can instantly send alerts or notifications to the homeowner's smartphone or computer. This real-time communication improves response time and allows users to take immediate action to protect their homes and valuable assets.

Furthermore, IoT-based security systems improve safety, automation, and convenience in everyday life. Homeowners can monitor live video feeds, control smart locks, and receive security updates from anywhere in the world. This technology also helps reduce human effort by automating many monitoring tasks and ensuring continuous surveillance without the need for manual supervision.

In addition, IoT-based smart home security systems contribute to the development of smart homes and intelligent living environments. They can be integrated with other home automation technologies such as smart lighting, fire detection systems, and emergency alarms to provide complete home protection. This integration enhances overall safety and creates a more secure and comfortable living environment for users.

Overall, the significance of IoT-based smart home security systems lies in their ability to provide real-time monitoring, automated protection, and remote accessibility. These systems not only improve home safety but also support the advancement of smart technologies that make everyday life more secure and convenient. As IoT technology continues to grow, smart home security systems will play an increasingly important role in protecting homes and enhancing modern lifestyles.

## 1.2 Objectives of IOT-Based Smart Home Security System

The main objective of this project, “IoT-Based Smart Home Security System” is to develop a smart and automated system that can monitor and protect homes using Internet of Things (IoT) technology. The project aims to design an efficient and reliable security solution that can detect suspicious activities and send real-time alerts to homeowners through internet-connected devices.

This system is designed to monitor different areas of a house using smart sensors, cameras, and microcontrollers. It focuses on applying IoT technology to improve home security and make monitoring easier through remote access. The system allows users to receive instant notifications and control security devices from anywhere using a smartphone or computer.

The specific objectives of this project are:

- To study and understand the basic concepts of the Internet of Things (IoT), smart devices, and home automation systems.
- To design and develop a smart home security system using IoT components such as sensors, cameras, and microcontrollers.
- To implement motion detection and intrusion detection using suitable sensors.
- To connect the system with the internet so that users can monitor their homes remotely.
- To create a system that can send real-time alerts or notifications when unusual activities are detected.
- To improve home safety by enabling automatic monitoring and quick response to potential security threats.
- To explore the use of IoT technology in modern smart homes and security applications.
- To understand how connected devices communicate and share data in an IoT-based system.
- To develop a reliable and efficient security model that can detect events such as motion, door opening, or unauthorized entry.
- To demonstrate how IoT technology can be used to create intelligent home security solutions for real-world applications.

### **1.3 Scope of IoT-Based Smart Home Security System**

The scope of this project, titled “IoT-Based Smart Home Security System” focuses on developing an intelligent system that can monitor and protect homes using Internet of Things (IoT) technology. The project explores how IoT devices, smart sensors, and internet connectivity can work together to improve home security and provide real-time monitoring. By connecting different devices such as motion sensors, door sensors, cameras, and alarms, the system allows homeowners to detect suspicious activities and receive instant alerts through their smartphones or computers.

This system is designed to detect unauthorized access, monitor movements, and ensure the safety of the home environment. The project covers the major aspects of a smart home security system, including the design of the IoT architecture, integration of sensors and microcontrollers, communication through the internet, and the monitoring of security events. The system continuously collects data from connected devices and provides notifications when unusual activities such as motion detection or door opening are detected.

The proposed IoT-based system can be applied in various real-world environments such as residential homes, apartments, offices, and smart buildings. It allows users to monitor their property remotely, improving safety and convenience. The system can also be expanded by integrating additional features such as fire detection, smart locks, and automated alarms to enhance overall home protection.

It demonstrates how IoT technology can transform conventional home security into a smart and connected system that improves monitoring, reduces human effort, and provides quick responses to potential threats. This project highlights the role of IoT in developing modern smart homes and intelligent security solutions for the future.

The scope of this project titled “IoT-Based Smart Home Security System” involves designing and implementing a smart monitoring system that can automatically detect security threats and notify users in real time. Using IoT technology and connected devices, the system can monitor different areas of the home and ensure continuous surveillance. The project also focuses on improving the efficiency, reliability, and convenience of home security through smart automation.

The scope further extends to practical applications in smart homes, residential buildings, and offices where automated security systems are required. This project demonstrates how IoT technology can help create intelligent security solutions that enhance safety, support remote monitoring, and contribute to the development of modern connected living environments.

## **Chapter 2: System Analysis**

### **2.1 Existing System:**

In the existing home security systems, monitoring and protection of houses are mainly carried out using traditional security devices such as alarms, CCTV cameras, and manual locks. These systems are usually standalone devices and do not provide real-time communication with the homeowner. Most traditional systems require constant human supervision to monitor the security cameras or respond to alarm signals, which makes them less efficient and time-consuming.

Traditional home security systems generally work by triggering an alarm when motion is detected or when a door or window is opened. However, these systems often lack smart connectivity and cannot send instant alerts to the user through the internet. In many cases, homeowners can only review recorded footage after an incident has already occurred, which reduces the effectiveness of the security system.

Another limitation of existing systems is that they are not easily accessible remotely. Many conventional security systems do not allow users to monitor their homes from distant locations. As a result, homeowners cannot check the status of their home security when they are away. These systems also have limited automation capabilities and cannot easily integrate with other smart devices.

In addition, traditional systems require manual installation and configuration, and they may involve high maintenance costs. They also have difficulty adapting to modern security needs such as real-time monitoring, remote alerts, and automated responses to suspicious activities.

Therefore, the existing home security systems have several limitations in terms of connectivity, automation, and real-time monitoring. To overcome these issues, a more advanced solution such as an IoT-based Smart Home Security System is required. Such a system can connect multiple devices through the internet, provide instant alerts, and allow homeowners to monitor and control their home security remotely, making the system more intelligent, efficient, and reliable.

## 2.2 Proposed System

The proposed system introduces an IoT-Based Smart Home Security System that uses smart sensors, internet connectivity, and automated alerts to provide better security and monitoring for homes. Unlike traditional security systems, the proposed system connects different devices such as motion sensors, door sensors, cameras, and alarms through the Internet of Things (IoT). This allows the system to monitor the home environment continuously and provide real-time updates to the homeowner.

In this system, sensors are installed at important locations such as doors, windows, and entry points of the house. These sensors can detect activities such as motion, door opening, or unusual movement. When any suspicious activity is detected, the system automatically sends notifications or alerts to the homeowner through a mobile application or web interface. This helps users take immediate action and monitor their homes from anywhere using the internet.

The system uses a microcontroller or IoT device to collect data from the connected sensors and transmit it to the cloud or server through a network connection. The user can view the status of the security system, receive alerts, and control certain devices remotely. This provides convenience and improves the efficiency of home security management.

The proposed system can be applied in different real-world environments such as residential homes, apartments, offices, and smart buildings. It reduces the need for constant manual monitoring and improves safety by providing automatic detection and real-time notifications. The system can also be expanded by integrating additional features such as smart locks, fire detection sensors, and surveillance cameras.

Unlike traditional systems that work independently, the IoT-based security system allows devices to communicate with each other and share data through the internet. This improves the reliability, flexibility, and scalability of the system. The system can be easily upgraded or expanded with additional devices as required.

The proposed system represents a modern and intelligent approach to home security. By combining IoT technology with smart sensors and automation, it provides a reliable solution for protecting homes and ensuring safety. The system enables continuous monitoring, remote access, and quick response to security threats, making it an effective solution for modern smart homes.

Furthermore, the proposed system aims to improve the overall performance and efficiency of traditional home security systems. It supports real-time monitoring, quick alert notifications, and remote control through connected devices. With the help of IoT technology and cloud connectivity, the system can handle large amounts of data and provide secure communication between devices.

Overall, the proposed IoT-based Smart Home Security System provides an intelligent, flexible, and scalable security solution. It helps homeowners monitor their property easily, improves safety, reduces human effort, and supports the development of modern smart homes and automated living environments.

In this system, various sensors such as motion sensors, door sensors, window sensors, and surveillance cameras are installed at important locations within the house. These devices continuously monitor the surrounding environment and detect unusual activities such as unauthorized entry, unexpected movement, or door opening. Once a suspicious event is detected, the system immediately sends alerts or notifications to the homeowner through a mobile application or web-based interface. This feature enables users to monitor their homes remotely and take necessary actions even when they are away.

The proposed system uses a microcontroller or IoT development board to connect and control all the sensors and devices. The microcontroller collects data from the sensors and processes it before sending it to a cloud platform or server through the internet. The cloud platform stores and manages the data, allowing the user to access real-time updates and security information at any time. This connectivity makes the system more efficient, flexible, and capable of handling multiple security events simultaneously.

Another important feature of the proposed system is remote monitoring and control. Homeowners can check the status of their security system from anywhere using smartphones, tablets, or computers. They can view live camera feeds, receive instant notifications, and control certain devices such as alarms or smart locks remotely. This remote accessibility increases convenience and ensures that the homeowner always remains aware of the security status of the house.

The proposed system also focuses on automation and quick response. When a sensor detects suspicious activity, the system can automatically trigger alarms, activate cameras, or send emergency notifications. This automated response reduces the need for constant human supervision and ensures faster reaction to potential security threats. The system can also be programmed to perform different actions depending on the type of event detected.

Furthermore, the system is scalable and flexible, which means it can easily be expanded by adding more sensors or smart devices in the future. For example, additional features such as fire detection sensors, gas leakage detectors, smart lighting, and smart door locks.

## 2.3 Requirement Analysis

Requirement analysis defines the necessary functionalities, performance expectations, and resources required to successfully develop the IoT-Based Smart Home Security System. It helps in identifying what the system should do, how it should perform, and the conditions under which it must operate. Proper requirement analysis ensures that all important functions, hardware components, and system resources are clearly defined for the successful implementation of the project.

### Functional Requirements:

- The system must be able to monitor the home environment using connected sensors.
- It should detect activities such as motion, door opening, or unauthorized entry using sensors.
- The system should automatically send alerts or notifications to the user when suspicious activity is detected.
- It should allow users to monitor the home remotely using a smartphone or web-based application.
- The system must be able to collect data from sensors, it through the using IoT technology.
- The system should trigger alarms or security notifications when abnormal events occur.
- It should provide real-time monitoring and status updates of connected security devices.
- The system must allow users to check the security status and receive alerts from anywhere.
- The system should support integration with multiple sensors such as motion sensors, door sensors, and surveillance cameras.
- It should provide a simple and easy-to-use interface for users to monitor and control the system.

### Sensor Monitoring Module:

The system must continuously collect data from sensors such as motion detectors, door sensors, and security cameras to detect unusual activities within the home environment.

### Alert and Notification System:

When the system detects suspicious activity, it should automatically send notifications or alerts to the homeowner through a mobile application, SMS, or internet-based messaging service.



### Non-Functional Requirements:

- It should have fast response time when detecting security events.
- The system should be scalable so that additional sensors or devices can be added easily.
- The interface must be simple, user-friendly, and easy to operate.
- The system should ensure secure communication between IoT devices and the user interface.
- It must use efficient memory and power resources to ensure stable operation.
- The system should work smoothly across different network environments.
- It should support integration with future smart home devices and automation systems.

### Performance:

The system should detect security events and send alerts with minimal delay to ensure quick response to threats.

### Scalability:

The architecture should allow easy addition of new sensors, cameras, or security devices without major system changes.

### Usability:

The system interface should be simple and accessible so that users with basic technical knowledge can operate it easily.

### Reliability:

The system should operate consistently and provide accurate monitoring results under normal operating conditions.

### Maintainability:

The system should have modular design and well-structured code so that future updates, maintenance, and upgrades can be performed easily.

### Internet Dependency:

The system relies on internet connectivity for communication and remote monitoring. Poor or unstable internet connections may affect real-time alerts.

### Hardware Limitations:

The performance of the system depends on the quality and capability of sensors, cameras, and microcontrollers used in the setup.

### Security Risks:

Since the system operates over the internet, there may be potential cybersecurity risks if proper data protection and encryption are not implemented.

## **Observed Limitations:**

Although the IoT-Based Smart Home Security System provides many advantages, some limitations may occur during implementation and operation. One major limitation is the dependency on internet connectivity.

Since the system relies on the internet to send alerts and enable remote monitoring, poor or unstable network connections can affect the performance of the system. Another limitation is hardware dependency, as the accuracy and reliability of the system depend on the quality of sensors and IoT devices used.

Environmental conditions can also affect sensor performance. For example, motion sensors may sometimes be triggered by pets, moving objects, or environmental changes, which may lead to false alarms.

In addition, the system requires continuous power supply to operate effectively, and power failures may temporarily disable the security system if backup power is not available.

There may also be security and privacy concerns if proper cybersecurity measures are not implemented, as IoT devices connected to the internet may become targets for unauthorized access.

Despite these limitations, the system remains an effective solution for improving home security and supporting modern smart home technology. Despite its advantages, the IoT-based smart home security system may have some limitations.

One major limitation is internet dependency, since the system relies on internet connectivity for sending alerts and enabling remote monitoring. If the network connection becomes unstable or unavailable, the system may not send notifications in real time.

Another limitation is hardware dependency, as the system performance depends on the quality of sensors, cameras, and microcontrollers used. Environmental conditions may also affect sensor accuracy. For example, motion sensors may sometimes detect non-threatening movements such as pets or moving objects, which can result in false alarms.

The system also requires a continuous power supply to function properly. Power failures can temporarily disable the system if backup power is not available.

In addition, since IoT devices are connected to the internet, there may be security and privacy risks if proper cybersecurity measures are not implemented. However, with proper system design and security practices, these limitations can be minimized while maintaining effective home protection.

## 2.4 Hardware Requirements

To ensure smooth operation and reliable monitoring of the IoT-Based Smart Home Security System, appropriate hardware components are required. These components help in detecting activities, processing data, and communicating information through the internet.

The following hardware specifications are recommended for implementing the smart home security system.

- Microcontroller / Development Board: Arduino Uno / Raspberry Pi / ESP8266 / ESP32
- Processor: Minimum 1 GHz processor (for Raspberry Pi based system)
- RAM: Minimum 2 GB RAM (4 GB recommended for better performance)
- Sensors: Motion Sensor (PIR), Door/Window Sensor
- Camera Module: USB Camera or Raspberry Pi Camera (optional for video monitoring)
- Storage: Minimum 32 GB SD Card or internal storage
- Power Supply: 5V / 12V power adapter for microcontroller and sensors
- Display (Optional): LCD display for showing system status
- Network Module: Wi-Fi module (ESP8266/ESP32) for internet connectivity
- Other Components: Buzzer/Alarm, Breadboard, Connecting Wires, Resistors

Hardware Components Table

| Component           | Specification                      | Description / Remarks                                                                         |
|---------------------|------------------------------------|-----------------------------------------------------------------------------------------------|
| Microcontroller     | Arduino Uno / Raspberry Pi / ESP32 | Acts as the central controller that processes sensor data and communicates with IoT platforms |
| Motion Sensor (PIR) | Standard PIR sensor module         | Detects human movement in the monitored area                                                  |
| Door Sensor         | Magnetic contact sensor            | Detects door or window opening                                                                |
| Wi-Fi Module        | ESP8266 / ESP32                    | Provides internet connectivity for IoT communication                                          |
| Alarm / Buzzer      | Standard buzzer module             | Produces alert sound when unauthorized activity is detected                                   |
| Other Components    | Breadboard, wires, resistors       | Used for circuit connections and system setup                                                 |

## 2.5 Software Requirements

The software environment plays an important role in developing and managing the IoT-based smart home security system. Software tools are used to program the microcontroller, monitor sensor data, and enable communication between devices and users.

The following software tools and technologies are required:

- Operating System: Windows 10 / 11 or Linux (Ubuntu)
- Programming Language: Python / C / C++
- IoT Platforms: Blynk / Thing Speak / Firebase / MQTT
- Libraries and Tools:
  - Arduino IDE libraries for sensor control
  - OpenCV (optional for camera monitoring)
  - NumPy and Pandas (for data handling if using Python)
    - Development Tools: Arduino IDE, Visual Studio Code, PyCharm
    - Database / Cloud Storage: Firebase / Thing Speak for storing sensor data
    - Communication Protocols: HTTP / MQTT / Wi-Fi communication

The software environment is designed to ensure smooth integration between sensors, microcontrollers, and cloud platforms. The Arduino IDE is commonly used for programming microcontrollers such as Arduino or ESP8266. Python can also be used when Raspberry Pi is used as the main controller.

IoT platforms such as Blynk or Thing Speak help in monitoring sensor data and sending alerts to users through mobile applications. These platforms provide dashboards where users can monitor system status in real time.

Development tools such as Visual Studio Code or PyCharm help in writing and debugging code efficiently. Cloud databases store sensor readings and security event logs, which can later be used for analysis and monitoring.

This software setup enables easy development, testing, and deployment of the IoT-based smart home security system.

## 2.6 Justification of platform

The platform selected for the development of the IoT-Based Smart Home Security System is chosen based on efficiency, reliability, scalability, and compatibility with IoT technologies.

Arduino, Raspberry Pi, or ESP-based microcontrollers are selected because they are cost-effective, easy to use, and widely supported for IoT applications. These devices can easily connect with sensors and communicate with internet platforms, making them suitable for smart home security solutions.

The Arduino IDE and Python programming language are used because they provide simple and flexible programming environments. They support a wide range of libraries that make sensor integration and IoT communication easier.

The use of IoT cloud platforms such as Blynk or Thing Speak allows real-time monitoring and remote control of the system. These platforms help users receive instant notifications when suspicious activities are detected.

The system is implemented on Windows or Linux operating systems, which provide compatibility with most IoT development tools and ensure stable performance.

The selected combination of microcontrollers, sensors, IoT platforms, and programming tools creates a reliable and efficient platform for building the smart home security system. It ensures real-time monitoring, fast communication, and easy scalability for future improvements.

Overall, the chosen platform provides a cost-effective, flexible, and scalable solution for implementing an IoT-based smart home security system suitable for both academic projects and real-world smart home applications.

The platform selected for the development of the project “IoT-Based Smart Home Security System” is carefully chosen to ensure efficiency, reliability, scalability, and compatibility with modern Internet of Things technologies. The integration of suitable hardware and software components plays an important role in building a system that can monitor the home environment effectively and provide real-time security alerts to users. The selected platform allows smooth communication between sensors, microcontrollers, cloud platforms, and user interfaces.

For the hardware platform, microcontrollers such as Arduino, Raspberry Pi, or ESP8266/ESP32 are used because they are widely recognized and commonly used in IoT-based projects. These devices are cost-effective, energy-efficient, and capable of connecting multiple sensors and communication modules. They act as the central control unit of the system by collecting data from various sensors such as motion sensors, door sensors, and cameras. After collecting the data, the microcontroller processes it and sends the information to the cloud platform or user device through internet connectivity.

The use of sensors such as PIR motion sensors, door or window sensors, and cameras further strengthens the functionality of the system. Motion sensors detect movement within the monitored area, while door or window sensors identify unauthorized access. Cameras can provide visual monitoring and record activities when suspicious movement is detected. The combination of these devices ensures accurate detection of security threats and provides reliable monitoring of the home environment.

For the software platform, Arduino IDE and Python programming language are selected because they provide flexible and user-friendly programming environments. Arduino IDE is widely used for programming microcontrollers and allows easy integration of sensors and communication modules. Python is also used in some cases, especially when Raspberry Pi is used as the main controller, because it supports many libraries that simplify hardware interaction and IoT communication. These programming environments make the system easy to develop, test, and maintain.

The project also uses IoT cloud platforms such as Blynk, Thing Speak, or Firebase for real-time monitoring and data management. These platforms allow the system to send sensor data to the cloud and display it on a mobile or web dashboard. Through these platforms, users can monitor the status of their home security system from anywhere in the world. They can receive instant notifications or alerts when suspicious activity is detected. Cloud platforms also store historical data, which can help users analyse security events or system performance in the future.

The development environment for this project is supported by operating systems such as Windows 10/11 or Linux (Ubuntu). These operating systems are chosen because they are compatible with most IoT development tools and provide a stable environment for programming and testing the system. Linux, in particular, offers better resource management, higher stability, and improved performance for IoT-based applications. It also supports various open-source tools and libraries that make development more efficient.

Another important reason for selecting this platform is its scalability and flexibility. The system can easily be expanded by adding additional sensors, cameras, or smart devices without making major changes to the existing system architecture. For example, the system can be extended to include fire detection sensors, gas leakage detectors, or smart door locks to provide a more advanced home automation and security solution.

The selected platform also provides cost-effectiveness, which makes it suitable for academic projects as well as real-world applications. Microcontrollers such as Arduino and ESP8266 are inexpensive and widely available, making the system affordable while still providing reliable performance. At the same time, cloud-based IoT platforms allow efficient data storage and remote monitoring without requiring complex infrastructure.

## Chapter 3: System Design

### 3.1 Sensor Monitoring Module

#### Responsibilities:

- Collect data from motion sensors, door sensors, and smoke sensors
- Detect abnormal activity
- Send signals to the controller

#### Components:

- PIR motion sensor
- Magnetic door sensor
- Gas/smoke sensor

### 3.2 Camera Surveillance Module

#### Responsibilities:

- Capture images or video when motion is detected
- Provide live video streaming
- Store images for later analysis

#### Features:

- Night vision support
- Motion detection recording
- Remote viewing

### 3.3 IoT Controller Module

#### Responsibilities:

- Connect sensors and cameras
- Process incoming data
- Trigger alarms if threats are detected

#### Hardware examples:

- Raspberry Pi
- ESP32
- Arduino

### 3.4 Alert & Notification Module

#### Responsibilities:

- Send notifications to the user when suspicious activity is detected

#### Alert types include:

- Mobile push notifications

- SMS alerts
- Alarm siren activation

### 3.5 Data Storage Module

This module stores system information such as:

- Sensor activity logs
- Images captured by cameras
- User information
- Security event history

Database options:

- MySQL
- Firebase
- MongoDB

### 3.6 Mobile / Web Application Module

Responsibilities:

- Provide user interface for monitoring the home
- Display live camera feed
- Show alerts and notifications
- Allow remote device control

Features include:

- Login authentication
- Activity history
- Device status monitoring
- Security settings

### 3.7 Security & Authentication Module

This module ensures system security.

Functions include:

- User login authentication
- Secure communication using encryption
- Access control for devices
- System activity logging



#### 4. Data Flow of the System

The working flow of the IoT Smart Home Security System is as follows:

1. Sensors continuously monitor the home environment.
2. When motion or unusual activity is detected, the sensor sends data to the IoT controller.
3. The controller processes the data and determines whether it is a threat.
4. If a threat is detected, the system activates the alarm and captures images through the camera.
5. The data is sent to the cloud server for storage.
6. The user receives a notification on the mobile application.
7. The user can view the live camera feed and take necessary action.

#### 5. Advantages of the System

- Provides real-time home security monitoring.
- Allows remote control and access.
- Improves safety using automated alerts.
- Scalable system that can add new devices easily.
- Cost-effective compared to traditional security systems.

#### 6. Hardware Components

The IoT-based smart home security system uses various hardware components to detect and monitor security threats.

##### 1. Raspberry Pi

Raspberry Pi acts as the main controller of the system. It collects data from sensors, processes it, and sends it to the cloud server. It also controls the camera and alarm system.

##### 2. PIR Motion Sensor

The PIR (Passive Infrared) sensor detects movement of people or animals by sensing infrared radiation. When motion is detected, it sends a signal to the controller.

##### 3. Camera Module

The camera captures images or videos when motion is detected. The captured data can be stored or sent to the user through the application.

##### 4. Buzzer / Alarm

The buzzer is used to generate an alarm sound when an unauthorized entry is detected.

## 5. Wi-Fi Module

The Wi-Fi module enables internet connectivity, allowing the system to send alerts and upload data to the cloud.

## 6. Door / Window Sensor

Magnetic sensors detect whether a door or window is opened unexpectedly.

## 7. Software Components

### 1. Programming Language

Python is commonly used to control the Raspberry Pi and process sensor data.

### 2. Database

The database stores system information such as:

- Sensor readings
- Security alerts
- User details
- Activity logs

Examples include MySQL, Firebase, or MongoDB.

### 3. Mobile / Web Application

The application allows users to:

- Monitor home security remotely
- View live camera feed
- Receive notifications
- Control devices

### 4. Cloud Platform

Cloud services store system data and provide remote access.

Examples include:

- AWS
- Google Firebase
- Microsoft Azure

## 8. System Workflow

The working process of the IoT-based smart home security system is as follows:

1. Sensors continuously monitor the home environment.
2. When motion or unusual activity is detected, the PIR sensor sends a signal to the Raspberry Pi.
3. The controller processes the signal and activates the camera.

## 9. Security Features

The system includes several security features to ensure safety:

- Real-time intrusion detection
- Instant mobile notifications
- Secure login authentication
- Encrypted communication between devices
- Automatic alarm activation

## 10. Advantages of IoT Smart Home Security System

- Provides 24/7 monitoring of the home
- Allows remote access and control
- Sends instant alerts to the user
- Improves home safety and security
- Easy to install and expand with more devices

## 11. Limitations

- Requires a stable internet connection
- Initial setup cost may be high
- IoT devices may face cybersecurity risks if not properly secured

## 12. Future Enhancements

The system can be improved in the future by adding:

- AI-based face recognition to identify family members and strangers
- Voice control integration with smart assistants
- Smart door lock system
- Automatic emergency service alerts
- Energy-efficient IoT sensors

### **3.1 Modular Division**

The proposed IoT-Based Smart Home Security System is divided into several modules to ensure a structured and efficient system architecture. Each module performs a specific task that contributes to the overall functionality of the security system. The modular approach improves system scalability, reliability, and ease of maintenance.

#### **1. Sensor Data Acquisition Module**

The Sensor Data Acquisition Module is the initial stage of the system. It is responsible for collecting environmental and security-related data from various IoT sensors installed in the smart home environment. These sensors include motion sensors, door and window sensors, smoke detectors, and gas sensors.

The module continuously monitors the surroundings and captures real-time data such as motion detection, door status, and environmental conditions. When any unusual activity is detected, the sensor sends signals to the central controller (such as Raspberry Pi or microcontroller). This module ensures that the system receives accurate and continuous input data for further processing.

#### **2. Data Processing and Preprocessing Module**

After the sensor data is collected, the Data Processing and Preprocessing Module processes and prepares the data for analysis. This module filters noise, validates sensor signals, and ensures that only meaningful and relevant data is passed to the next stage.

For camera-based monitoring, the module may perform image preprocessing operations such as resizing, noise reduction, and brightness adjustment. These preprocessing steps improve the quality of captured images and make them suitable for further analysis or detection processes.

#### **3. Intrusion Detection Module**

The Intrusion Detection Module is responsible for identifying suspicious or unauthorized activities within the monitored environment. It analyzes the data received from sensors and cameras to determine whether a potential security threat exists.

For example, if a motion sensor detects movement when the house is supposed to be empty or a door sensor detects unauthorized entry, the system interprets this as a possible intrusion. Advanced implementations may also use artificial intelligence or image recognition techniques to distinguish between humans, pets, or harmless movements.

#### 4. Alert and Notification Module

The Alert and Notification Module is responsible for informing the homeowner or authorized user when suspicious activity is detected. Once an intrusion or abnormal event is confirmed, the system generates alerts.

Notifications can be sent through various communication channels, including:

- Mobile push notifications
- SMS alerts
- Email notifications
- Alarm or buzzer activation

This module ensures that the user receives real-time alerts so that appropriate action can be taken immediately.

#### 5. Data Storage and Cloud Integration Module

The Data Storage Module stores important system information for future reference and analysis. It records sensor logs, event history, images captured by cameras, and system status information.

Cloud platforms or databases are used to store this information securely. Storing data in the cloud allows the user to access system logs and security footage remotely through a web or mobile application. This module ensures reliable data management and accessibility.

#### 6. User Interface and Remote Monitoring Module

The User Interface Module provides an interactive platform for users to monitor and control their smart home security system. It can be implemented through a mobile application or web-based dashboard.

Through this interface, users can:

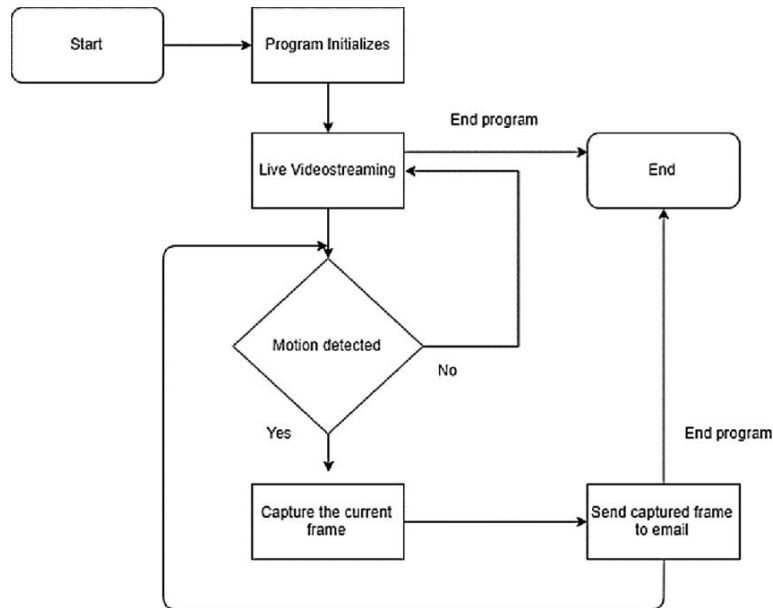
- View live camera feeds
- Receive and manage alerts
- Monitor sensor activity
- Control security devices remotely
- Access past security logs

This module enhances the usability of the system by allowing users to interact with the system easily from any location.

## 7. System Integration and Performance Monitoring Module

The System Integration and Performance Monitoring Module ensures smooth communication between all system components. It manages the data flow between sensors, controllers, cloud services, and user interfaces.

System Flow Diagram:



## 3.2 Data Dictionary

The data dictionary defines all the important data elements used in the IoT-Based Smart Home Security System, including their names, data types, and purposes. It helps maintain consistency, accuracy, and clarity while designing and implementing the system. The data dictionary ensures that all developers and users understand how information is stored, processed, and transmitted within the security system. It also helps organize data collected from sensors and system components in a structured manner.

Table: Data Dictionary

| Field Name    | Description                          | Data Type | Size / Format       | Remarks            |
|---------------|--------------------------------------|-----------|---------------------|--------------------|
| Device_ID     | Unique identifier of IoT device      | Integer   | 5 digits            | Primary key        |
| Sensor_ID     | Unique identifier for sensor         | Integer   | 5 digits            | Sensor reference   |
| Motion_Status | Status of PIR motion sensor          | Boolean   | True / False        | Motion detected    |
| Door_Status   | Status of magnetic door sensor       | Boolean   | Open / Close        | Door detection     |
| Alert_Status  | Indicates if alert is triggered      | Boolean   | True / False        | Security alert     |
| Event_Type    | Type of detected event               | String    | 50 chars            | Motion or Door     |
| Timestamp     | Date and time of event               | DateTime  | DD-MM-YYYY<br>HH:MM | Event record       |
| User_ID       | Identifier of user monitoring system | Integer   | 5 digits            | User reference     |
| Device_Status | System operational status            | String    | Active / Inactive   | Monitoring state   |
| Response_Time | Time taken to send alert             | Float     | Seconds             | Performance metric |

The data dictionary for the project IoT-Based Smart Home Security System defines the structure and meaning of the data elements used throughout the system. Each IoT device used in the system is uniquely identified by a Device\_ID, which helps in distinguishing different devices connected within the security network.

The Motion\_Status field records whether motion has been detected by the PIR sensor. It usually stores values such as *true* or *false* depending on the sensor output. Similarly, the Door\_Status field records whether the door is open or closed based on the magnetic door sensor reading. These fields are essential for determining whether any suspicious activity has occurred within the home environment.

When any abnormal event occurs, the Alert\_Status field indicates whether the system has triggered a security alert. The Event\_Type field identifies the type of event detected by the system, such as motion detection or door opening. This information helps users quickly understand the nature of the security event.

The Timestamp field stores the exact date and time when the event occurred. This ensures proper chronological tracking of all security activities and helps in reviewing system logs. The User\_ID field represents the user who is monitoring the system or receiving the alert notifications. This is useful in systems where multiple users may access the security dashboard.

The Device\_Status field shows whether the IoT device is currently active or inactive. This helps monitor the operational status of the system and ensures that all components are functioning correctly. The Response\_Time field records the time taken by the system to send alerts or notifications after detecting an event. This metric helps evaluate system performance and efficiency.

The data dictionary ensures that all data collected from sensors and devices are stored in a structured and consistent format. This helps developers maintain data accuracy and makes system integration easier.

- System Modules of IoT-Based Smart Home Security System

The proposed IoT-Based Smart Home Security System is designed using several interconnected modules that work together to monitor home security, detect suspicious activities, and notify users in real time. Each module performs a specific function and contributes to the overall efficiency of the system.

1. Sensor Data Acquisition Module

This module acts as the entry point of the system. Its main function is to collect data from different sensors installed in the home environment. Sensors such as PIR motion sensors and magnetic door sensors continuously monitor their surroundings. The PIR sensor detects movement within the monitored area, while the magnetic sensor detects whether a door or window is opened or closed.



The collected data is sent to the microcontroller for further processing. This module ensures that all sensor readings are captured accurately and in real time.

Purpose: To collect security-related data from sensors.

Input: Motion signals and door status data.

Output: Raw sensor data sent to the processing unit.

## 2. Data Processing and Control Module

The NodeMCU microcontroller acts as the central controller in this module. It receives sensor data and processes it according to predefined conditions programmed in the system. If motion or unauthorized door opening is detected, the microcontroller identifies it as a potential security threat.

The microcontroller also manages communication between sensors, alarms, and cloud platforms. It ensures that sensor data is processed quickly and appropriate actions are triggered.

Purpose: To analyze sensor inputs and control system responses.

Input: Raw data from sensors.

Output: Processed data and event detection results.

## 3. Alert and Notification Module

This module is responsible for sending alerts to users when suspicious activities are detected. When the system identifies an abnormal event, it activates the buzzer alarm and sends notifications through the IoT platform or mobile application.

The notification may include information such as the type of event detected, the time of occurrence, and the sensor responsible for detection. This allows homeowners to take immediate action when a potential security threat occurs.

Purpose: To notify users about security threats.

Output: Alarm activation and user notifications.

## 4. IoT Communication Module

The IoT communication module enables the system to connect with internet platforms and mobile applications. The NodeMCU Wi-Fi module sends sensor data to a cloud server or IoT dashboard where it can be monitored remotely.

This module allows users to view real-time system updates, receive alerts, and monitor their home security system from anywhere using the internet.

Purpose: To enable remote monitoring and communication.

Output: Real-time data transmission to IoT platform.

## 5. Event Logging and Data Storage Module

This module records all security events detected by the system. Information such as motion detection, door opening, timestamps, and alert status is stored in a database or cloud storage. Maintaining event logs helps users review past security activities and analyze system performance.

These records can also be used to improve the system or detect patterns of suspicious activities.

Purpose: To store security event data.

Output: Organized database of system logs.

## 6. Alarm and Response Module

The alarm module is responsible for triggering a buzzer or alarm whenever a security breach is detected. This immediate response alerts people nearby and discourages unauthorized entry.

The buzzer is activated by the microcontroller when sensor conditions indicate suspicious activity.

Purpose: To provide immediate warning during security threats.

Output: Audible alarm signals.

## 7. System Monitoring and Maintenance Module

This module ensures that all components of the security system operate properly. It monitors device status, sensor functionality, and network connectivity.

If any component stops working or loses connection, the system can notify the user or technician for maintenance.

Purpose: To maintain system reliability and operational stability.

Output: System health status and maintenance alerts.

### 3.3 ER Diagram

The Entity Relationship (ER) Diagram represents the logical structure of the IoT-Based Smart Home Security System and illustrates how different entities in the system are related to each other. It helps in understanding how data is stored, connected, and managed within the system database. The ER diagram provides a clear visualization of entities such as users, sensors, devices, and security events along with their relationships.

#### Entities

1. User :

The user represents the homeowner or authorized person who interacts with the smart home security system. The user can monitor security alerts, receive notifications, and manage system settings through a mobile application or IoT dashboard.

2. Device :

The device entity represents the hardware component used in the system, mainly the NodeMCU microcontroller. It acts as the central controller that receives data from sensors and processes it. Each device has a unique device ID that helps identify it within the IoT network.

3. Sensor:

Sensors are responsible for detecting activities within the home environment. In this system, sensors include the PIR motion sensor and the magnetic door sensor. These sensors send signals to the NodeMCU whenever motion is detected or when the door is opened.

4. Event :

The event entity represents a security activity detected by the sensors. Events may include motion detection, door opening, or any suspicious activity recorded by the system. Each event stores details such as event type, time, and the sensor that triggered it.

5. Alert :

The alert entity stores information related to security notifications generated by the system. When an event indicates a possible security breach, the system triggers an alert through a buzzer or mobile notification to inform the user.

#### Relationships

- A User can manage multiple Devices.

This means one user can monitor and control several IoT devices installed in different parts of the house.

- Each Device is connected to multiple Sensors.

The NodeMCU device receives input from multiple sensors such as motion sensors and door sensors.

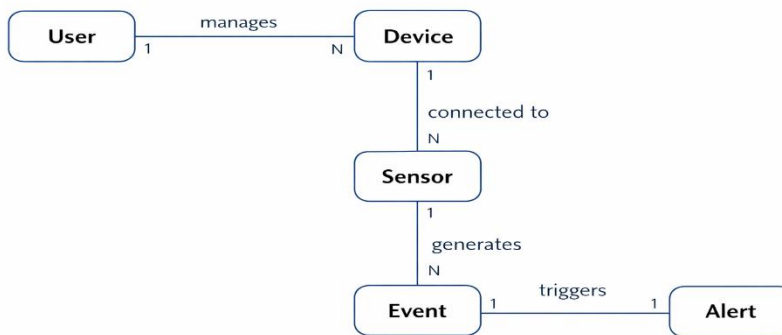
- Each Sensor generates multiple Events.

Whenever the sensor detects motion or door activity, an event record is created in the system.

- Each Event triggers one Alert.

If an event indicates suspicious activity, the system generates an alert that notifies the user and activates the buzzer alarm.

### ER Diagram: IoT-Based Smart Home Security System



### 3.4 Data Flow Diagram

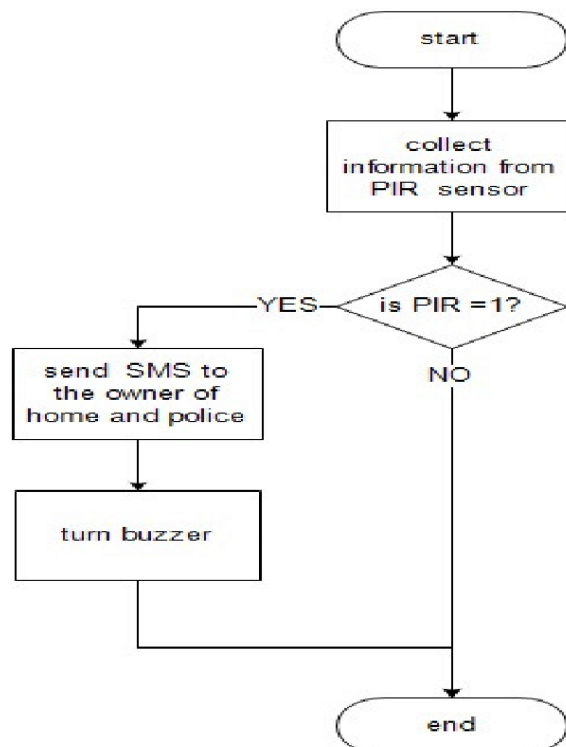
A Data Flow Diagram (DFD) represents how information moves through the system. It shows how data enters the system, how it is processed, and how the final output is generated. DFD helps in understanding the interaction between sensors, microcontroller, and alert systems in the IoT Based Smart Home Security System.

#### 3.4.1 System Flow Diagram

The system flow begins when the sensors monitor the home environment. The PIR motion sensor detects movement while the magnetic door sensor checks whether the door is open or closed. The sensor data is sent to the NodeMCU microcontroller, which processes the information. If any suspicious activity is detected, the system activates the buzzer alarm and sends alerts to the user through the IoT network.

##### Flow Steps

1. Sensors continuously monitor the home environment.
2. PIR sensor detects motion and magnetic sensor detects door status.
3. Sensor data is sent to the NodeMCU microcontroller.
4. NodeMCU processes the data and checks security conditions.
5. If intrusion is detected, the buzzer alarm is activated.
6. The system sends a notification alert to the user.
7. Event data is stored for monitoring and future analysis.



### 3.4.2 Class Diagram

The Class Diagram represents the structure of system components and their relationships in an object-oriented manner. Each class represents a part of the IoT security system and defines its responsibilities.

#### Main Classes

**SensorManager** – Handles data collection from PIR motion sensor and magnetic door sensor.

**DeviceController** – Represents the NodeMCU microcontroller which processes sensor data and controls system operations.

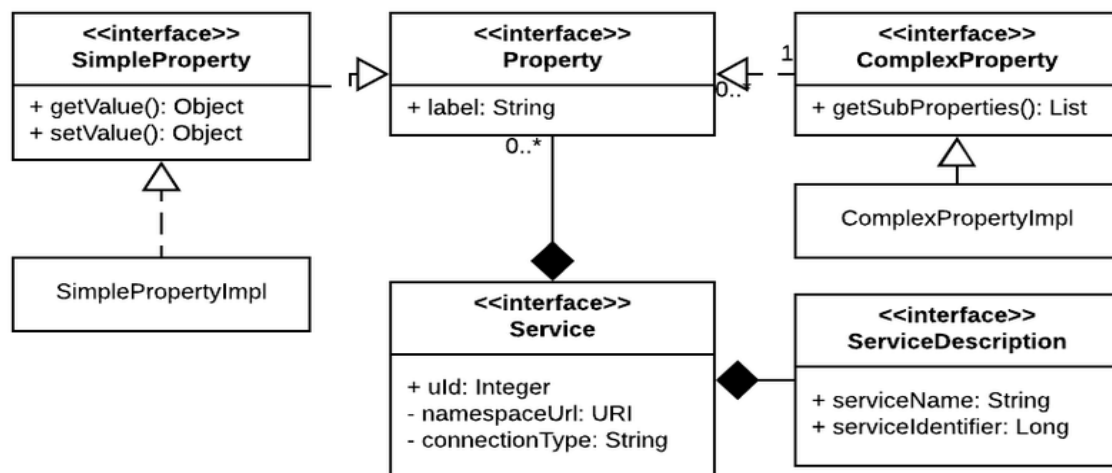
**AlertSystem** – Responsible for activating the buzzer and sending notifications when a security breach is detected.

**UserInfo** – Allows users to monitor system status and receive alerts through a mobile or IoT dashboard.

**DataLogger** – Stores event records such as motion detection, door opening, and alert notifications.

#### Relationships

- SensorManager → sends sensor data → DeviceController
- DeviceController → processes data → AlertSystem
- AlertSystem → sends alerts → UserInfo
- UserInfo → updates information → DataLogger

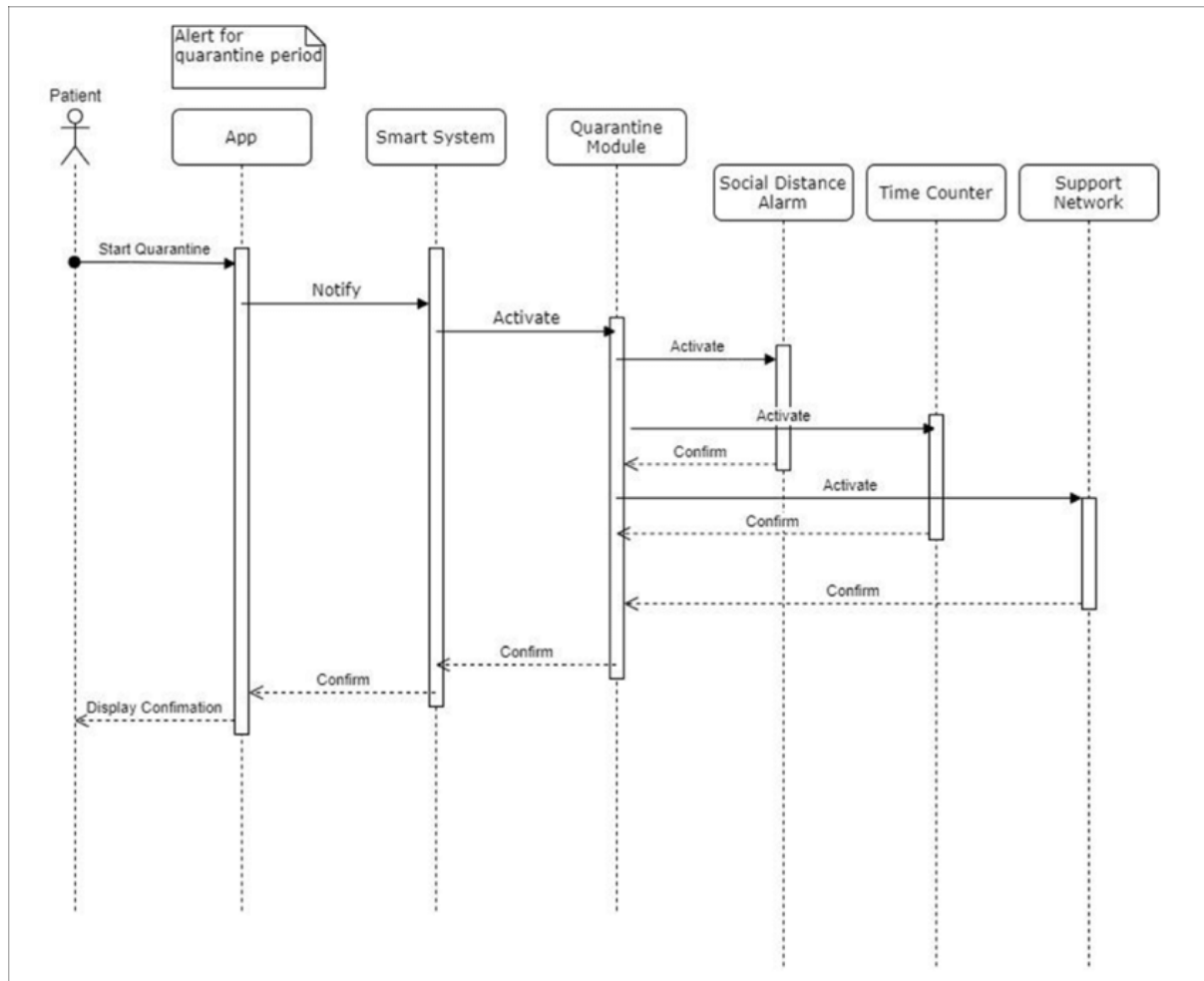


### 3.4.3 Sequence Diagram

The Sequence Diagram describes how different components of the system interact over time to detect and respond to security events.

#### Sequence

1. Sensors continuously monitor the environment.
2. When motion or door activity is detected, the sensor sends a signal to NodeMCU.
3. NodeMCU processes the received sensor data.
4. If suspicious activity is detected, NodeMCU activates the buzzer alarm.
5. The system sends an alert notification to the user.
6. The event information is recorded in the system database or IoT platform.



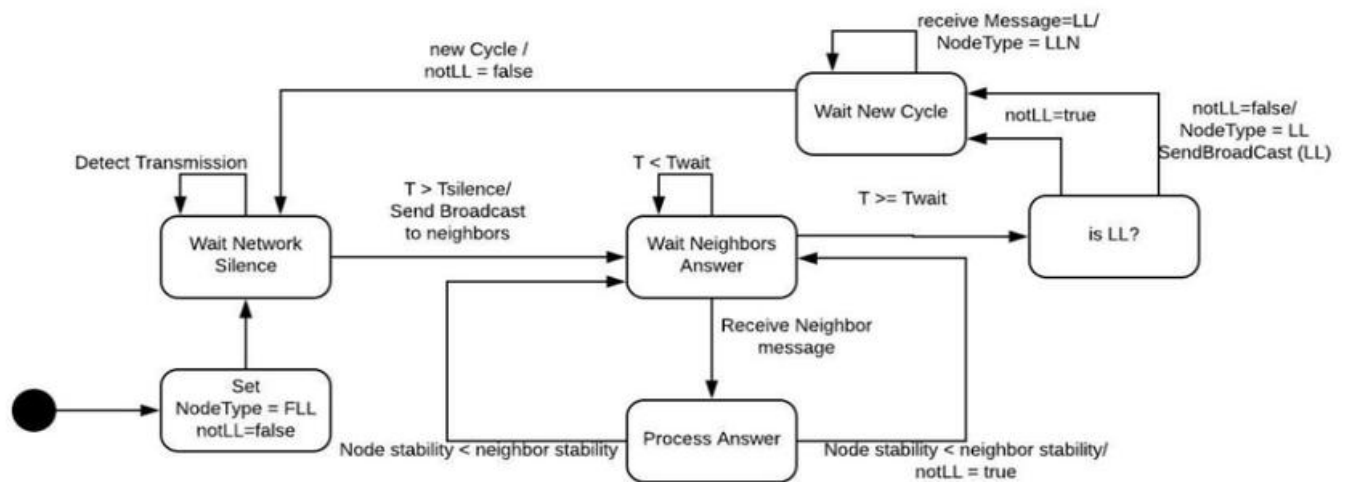
### 3.4.4 State Chart Diagram

The State Chart Diagram shows how the system changes states based on sensor inputs and system operations.

#### States

1. Idle State – System is active and waiting for sensor input.
2. Monitoring State – Sensors continuously monitor motion and door activity.
3. Detection State – Motion or door opening is detected by sensors.
4. Processing State – NodeMCU processes the sensor data.
5. Alert State – Buzzer alarm is activated and user notification is sent.
6. Event Logged State – Security event is stored in the system for future reference.

Transitions occur sequentially when sensors detect changes and the system responds accordingly.





### 3.4.5 Use Case Diagram

The Use Case Diagram shows how the user interacts with the IoT Smart Home Security System and how the system performs security operations.

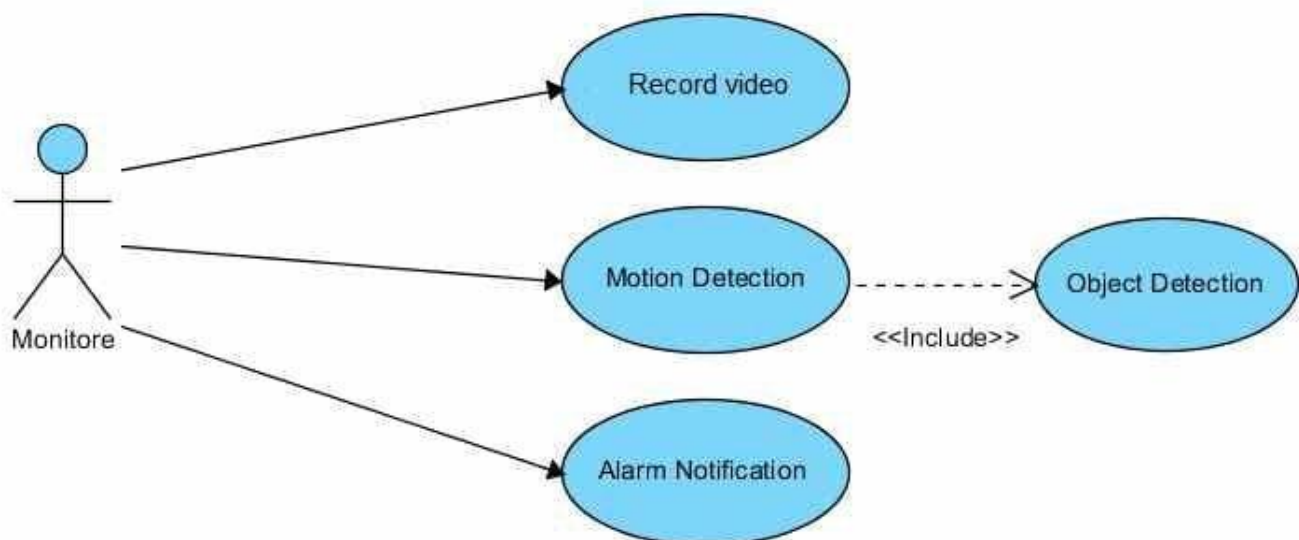
Actors

User – The homeowner or authorized person who monitors the system and receives alerts.

System – The IoT security system consisting of NodeMCU, sensors, and alarm components.

Use Cases

1. Monitor home environment
2. Detect motion activity
3. Detect door opening or closing
4. Trigger alarm and notifications
5. View security alerts and system status



## Chapter 4: Implementation and Testing

### 4.1 Code:

```
sketch_feb24a | Arduino IDE 2.3.7
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...

sketch_feb24a.ino
1  #define BLYNK_TEMPLATE_ID "TMPL3apk71-it"
2  #define BLYNK_TEMPLATE_NAME "Home Security System"
3  #define BLYNK_AUTH_TOKEN "OZqpc1rO_9vITYdmlEZe_j0dzYUs16L6"
4
5  #include <ESP8266WiFi.h>
6  #include <BlynkSimpleEsp8266.h>
7
8  char ssid[] = "Dharmi";
9  char pass[] = "123456789";
10
11 #define PIR_PIN 14
12 #define DOOR_PIN 12
13 #define BUZZER 13
14
15 BlynkTimer timer;
16
17 bool motionTriggered = false;
18 bool doorTriggered = false;
19
20 // ♦ This runs when ESP connects to Blynk server
21 BLYNK_CONNECTED()
22 {
23   Serial.println("✅ Connected to Blynk Server");
24   Blynk.syncAll(); // Sync virtual pins
25   Blynk.logEvent("device_status", "✅ ESP8266 Connected to Blynk Server");
26 }
27
28 void checkConnection()
29 {
30   if (Blynk.connected())
31   {
32     Serial.println("🟢 Blynk Connected");
```

```
sketch_feb24a | Arduino IDE 2.3.7
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...

sketch_feb24a.ino
33   }
34   else
35   {
36     Serial.println("🔴 Blynk Disconnected");
37   }
38 }
39
40 void checkSensors()
41 {
42   int motion = digitalRead(PIR_PIN);
43   int door = digitalRead(DOOR_PIN);
44
45   // MOTION SENSOR
46   if (motion == LOW)
47   {
48     Serial.println("⚠ Motion Detected");
49     Blynk.virtualWrite(V0, 1);
50     Serial.println("🔊 Motion status sent to Blynk");
51     digitalWrite(BUZZER, HIGH);
52
53     if (!motionTriggered)
54     {
55       Blynk.logEvent("motion_alert", "⚠ Motion detected in your home!");
56       motionTriggered = true;
57     }
58   }
59   else
60   {
61     Blynk.virtualWrite(V0, 0);
62     digitalWrite(BUZZER, LOW);
63     motionTriggered = false;
64   }
```

sketch\_feb24a.ino

```
65
66 // DOOR SENSOR
67 if (door == HIGH)
68 {
69   Serial.println("■ Door Opened");
70   Blynk.virtualWrite(V1, 1);
71   Serial.println("🔴 Door status sent to Blynk");
72   digitalWrite(BUZZER, HIGH);
73
74   if (!doorTriggered)
75   {
76     Blynk.logEvent("door_alert", "■ Door opened!");
77     doorTriggered = true;
78   }
79 }
80 else
81 {
82   Serial.println("Door Closed");
83   Blynk.virtualWrite(V1, 0);
84   doorTriggered = false;
85 }
86 }
87
88 void setup()
89 {
90   Serial.begin(9600);
91
92   pinMode(PIR_PIN, INPUT);
93   pinMode(DOOR_PIN, INPUT_PULLUP);
94   pinMode(BUZZER, OUTPUT);
95
96   Serial.println("📶 Connecting to WiFi & Blynk...");
```

```
97   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
98
99   timer.setInterval(1000L, checkSensors);
100   timer.setInterval(5000L, checkConnection); // Check connection every 5 sec
101 }
102
103 void loop()
104 {
105   Blynk.run();
106   timer.run();
107 }
```

## 4.2 Testing Approach

The testing approach for the IoT-Based Smart Home Security System is an important stage that ensures the system functions correctly, reliably, and efficiently in real-world situations. Since the project integrates IoT devices, sensors, and a microcontroller to monitor home security, a structured testing strategy is required to verify that all hardware and software components operate as expected.

Testing begins after the development and implementation of all system modules are completed. The main goal of testing is to ensure that the system can accurately detect motion or door activity and respond by triggering alerts and alarms. The testing process also ensures that communication between sensors, the NodeMCU microcontroller, and the alert system works smoothly.

The testing process follows multiple levels such as unit testing, integration testing, system testing, and performance testing, which help evaluate different aspects of the system. Unit Testing focuses on verifying the functionality of individual components used in the system. Each hardware component such as the PIR motion sensor, magnetic door sensor, NodeMCU microcontroller, and buzzer is tested separately. For example, the PIR sensor is tested to confirm that it can accurately detect motion, while the magnetic door sensor is tested to ensure it properly identifies door opening and closing conditions.

Integration Testing ensures that all components work together correctly. During this stage, the interaction between sensors and the NodeMCU controller is tested. The system verifies that when motion or door activity is detected, the signal is correctly transmitted to the microcontroller, which then processes the data and activates the buzzer alarm.

System Testing evaluates the complete system to confirm that it performs according to the project requirements. In this stage, the entire security system is tested in a real environment. Sensors monitor the surroundings, the NodeMCU processes incoming signals, and alerts are generated whenever suspicious activity occurs.

Performance Testing measures the efficiency and reliability of the system. This includes checking how quickly the system responds to sensor inputs, how accurately it detects motion or door activity, and whether the buzzer alarm is activated without delay. Performance testing also ensures that the system operates continuously without failures.

In addition to these testing methods, real-world scenario testing is also performed. The system is tested in different situations such as low light conditions, normal home activities, and door opening events to ensure that the sensors respond correctly. These tests help evaluate the reliability of the system in practical environments.

Error detection and troubleshooting are also important parts of the testing phase. If the system fails to detect motion or triggers false alarms, the issue is analyzed and corrected by adjusting sensor sensitivity, improving wiring connections, or modifying the NodeMCU program.

Finally, testing ensures that the system is stable, reliable, and efficient for home security monitoring. The testing process also confirms that the system can operate continuously and provide timely alerts whenever security threats are detected.

In conclusion, the testing approach for the IoT-Based Smart Home Security System ensures that all hardware and software components are thoroughly evaluated. This comprehensive testing strategy guarantees that the system functions effectively, providing a reliable and efficient solution for smart home security.

## **Chapter 5: Results and Discussions**

This chapter presents the results and discussion of the IoT-Based Smart Home Security System in terms of its performance, reliability, and effectiveness. The purpose of this chapter is to evaluate how well the system performs in detecting motion and door activity, triggering alerts, and ensuring home security. The testing and evaluation were conducted under different scenarios to verify that the system functions properly and meets the objectives of the project.

The chapter also discusses both quantitative and qualitative aspects of the system. Quantitative evaluation focuses on measurable factors such as detection accuracy, response time, and system reliability. Qualitative evaluation focuses on usability, ease of operation, and overall system effectiveness.

Furthermore, this chapter examines the stability and reliability of the hardware components such as the PIR motion sensor, magnetic door sensor, NodeMCU microcontroller, and buzzer alarm. The discussion highlights how well the system responds to potential security threats and how effectively it alerts the user. The results demonstrate that the system provides a reliable and efficient solution for monitoring home security using IoT technology.

### **5.1 Functionality Evaluation**

The functionality evaluation focuses on determining how accurately and efficiently the IoT-Based Smart Home Security System performs its intended tasks. The main objective of the system is to detect suspicious activities such as motion detection or unauthorized door opening and immediately alert the user.

The system was tested in different situations to evaluate its performance. These scenarios included normal home conditions, motion detection within the sensor range, and opening or closing of doors using the magnetic door sensor. The tests were conducted multiple times to verify the accuracy and consistency of the system.

- The evaluation mainly focused on several important performance factors:

#### **Detection Accuracy:**

The PIR motion sensor was tested to determine how accurately it detects human movement within the monitored area. The magnetic door sensor was also tested to ensure that it correctly identifies whether a door is opened or closed.

#### Response Time:

The response time measures how quickly the system reacts after detecting motion or door activity. Once the sensor detects an event, the NodeMCU processes the signal and activates the buzzer alarm within a short period of time.

#### System Reliability:

Reliability testing ensures that the system continues to function properly without errors or interruptions. During testing, the sensors continuously monitored the environment, and the system successfully detected security events without major failures.

#### Alert Mechanism:

The buzzer alarm was tested to verify that it activates whenever suspicious activity is detected. This immediate alert helps notify people nearby and provides an effective warning mechanism.

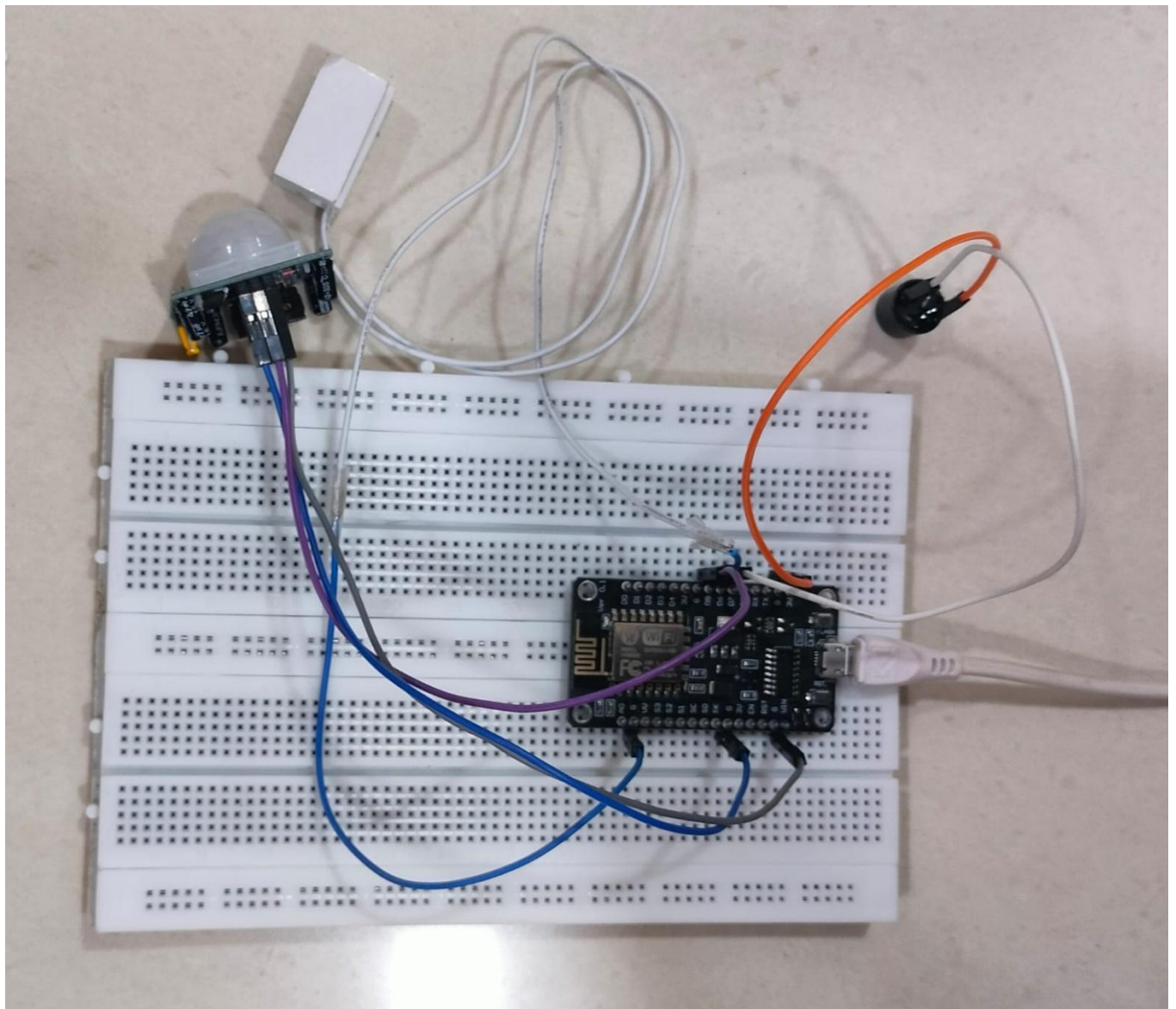
#### Integration Performance:

The interaction between hardware components such as the PIR sensor, magnetic door sensor, NodeMCU microcontroller, and buzzer was evaluated to ensure that all components communicate properly.

The results confirmed that the sensors successfully transmit signals to the microcontroller, which processes the information and activates the alarm system.

The results from the functionality evaluation show that the IoT-Based Smart Home Security System performs effectively and reliably.

The sensors accurately detect motion and door activity, the NodeMCU processes the data quickly, and the buzzer alarm provides immediate alerts when a potential security threat is detected.





## 5.2 User Experience Assessment

The User Experience (UX) Assessment for the IoT-Based Smart Home Security System was conducted to evaluate how easily and effectively users can interact with the system and understand the security alerts generated by it. The main objective of this assessment was to measure usability, accessibility, system responsiveness, and overall user satisfaction when using the smart home security setup.

Key UX parameters such as ease of installation, clarity of alerts, system responsiveness, reliability of notifications, and overall satisfaction were evaluated. Since the system is designed to automatically monitor home environments using sensors and a microcontroller, the user mainly interacts with the system through alert notifications and the buzzer alarm.

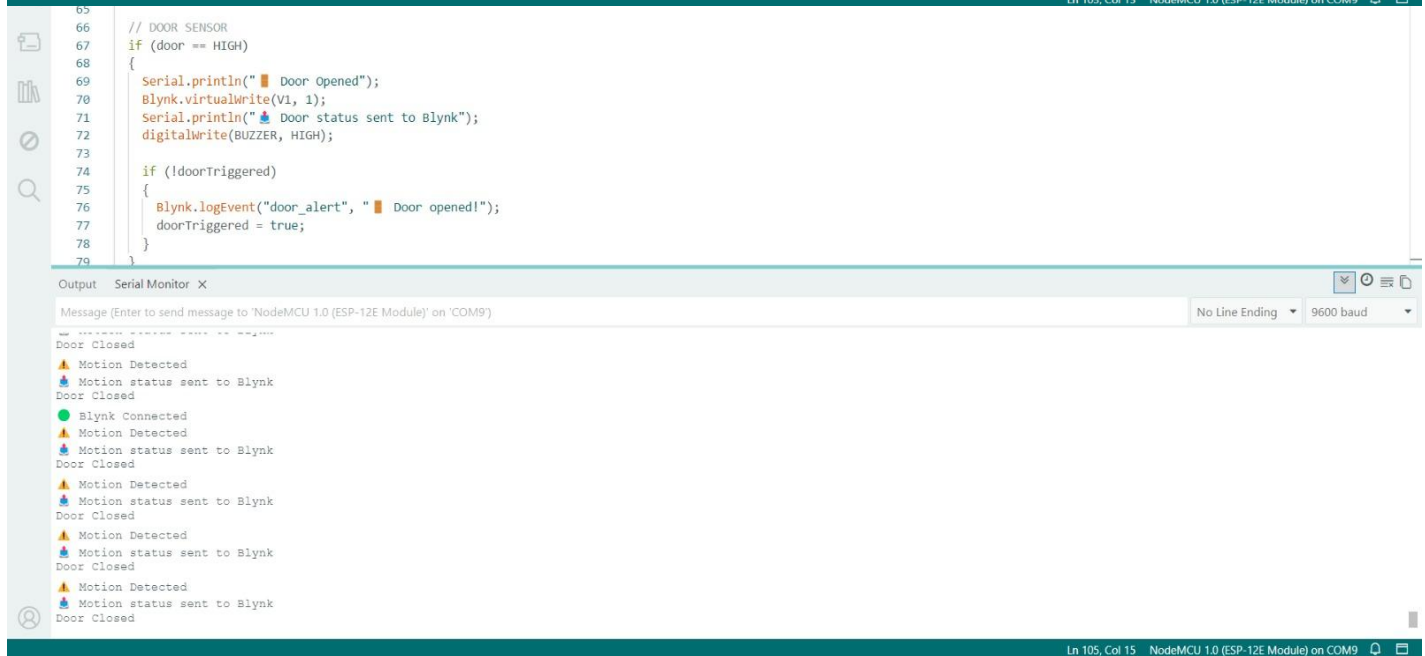
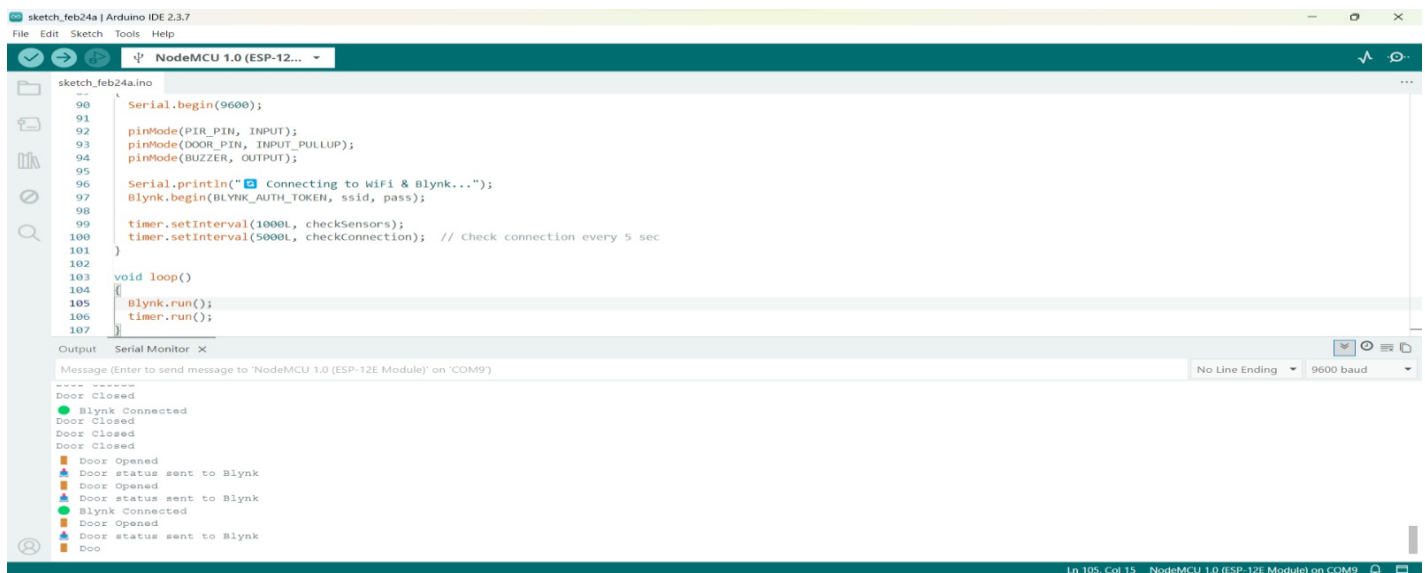
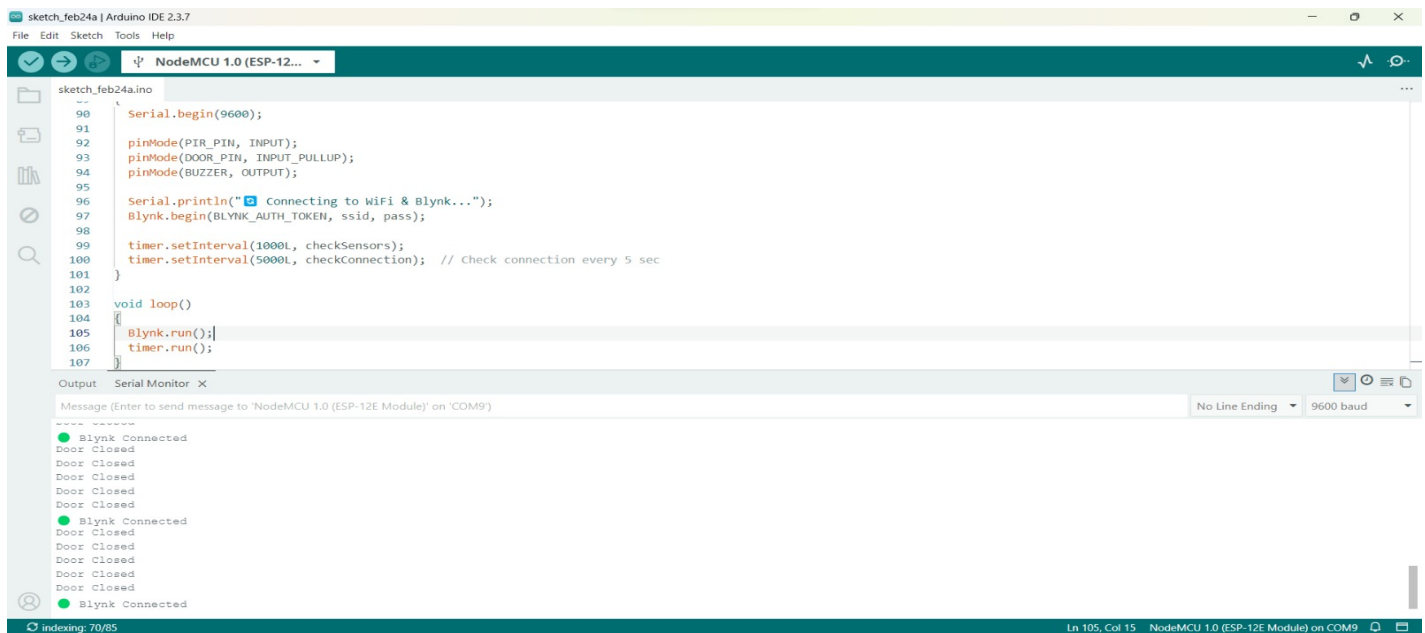
During testing, users observed how the system behaves when motion is detected or when the door is opened. Participants evaluated how quickly the system responds to security events and how clearly the alerts are communicated. The buzzer alarm and notifications were checked to ensure that they effectively notify users about potential security threats.

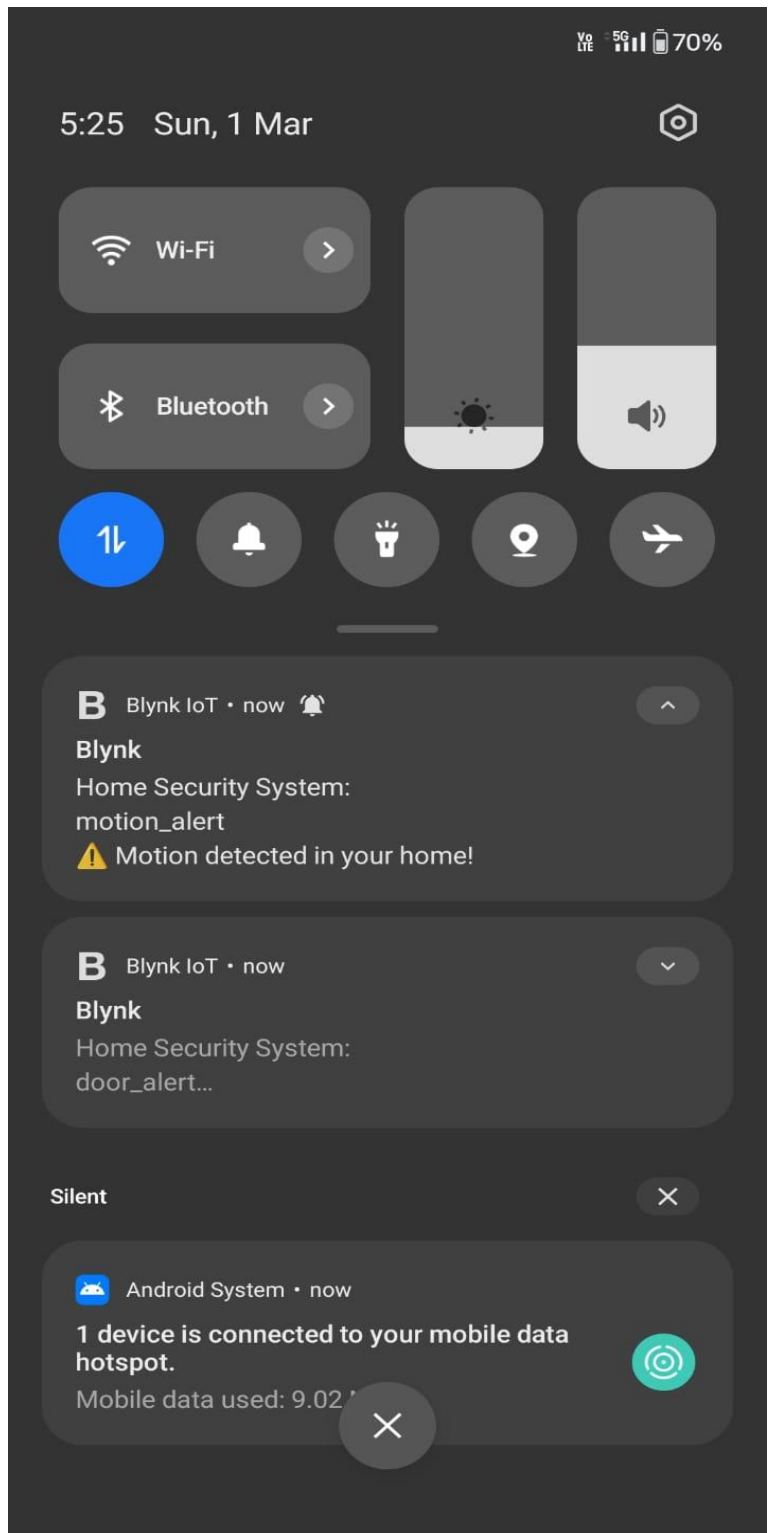
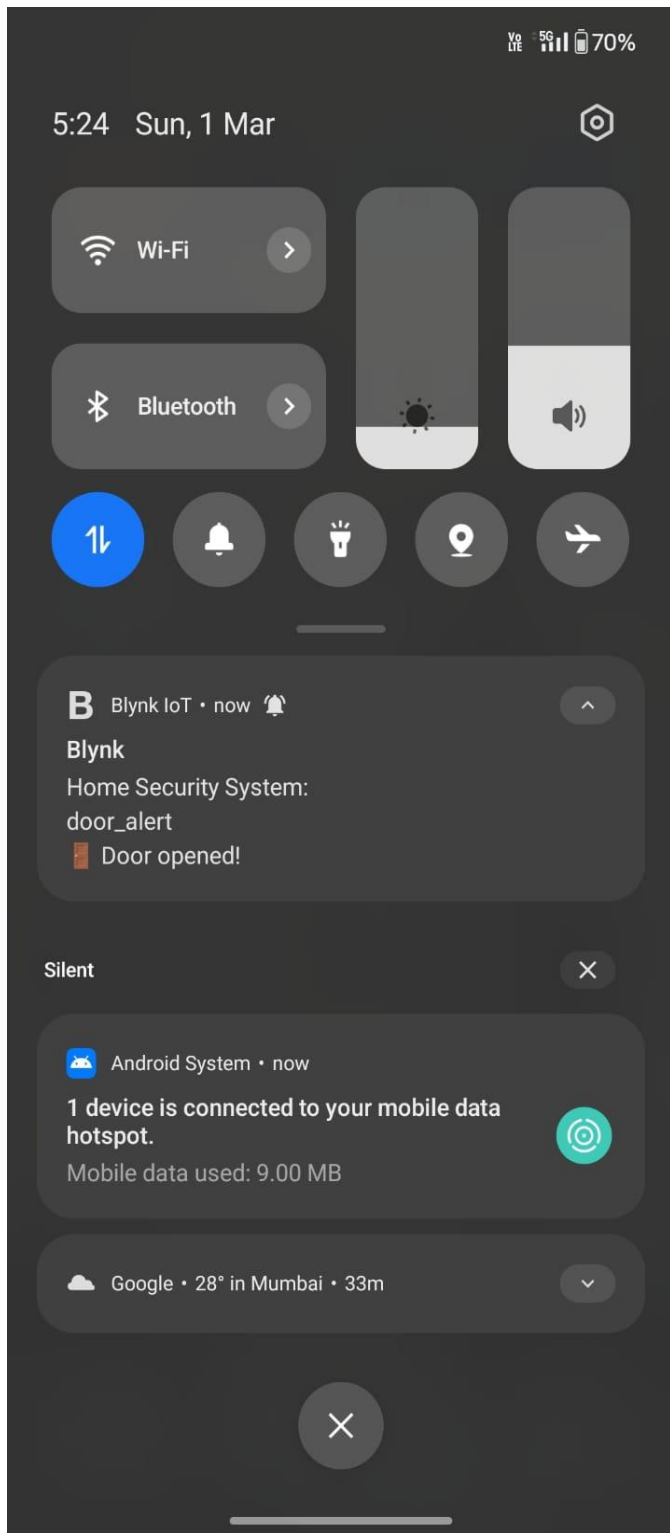
The system interface and hardware setup were also evaluated for simplicity and accessibility. Users were able to understand the functioning of the system easily, even with minimal technical knowledge. The setup using NodeMCU, PIR motion sensor, magnetic door sensor, and buzzer was found to be straightforward and easy to operate.

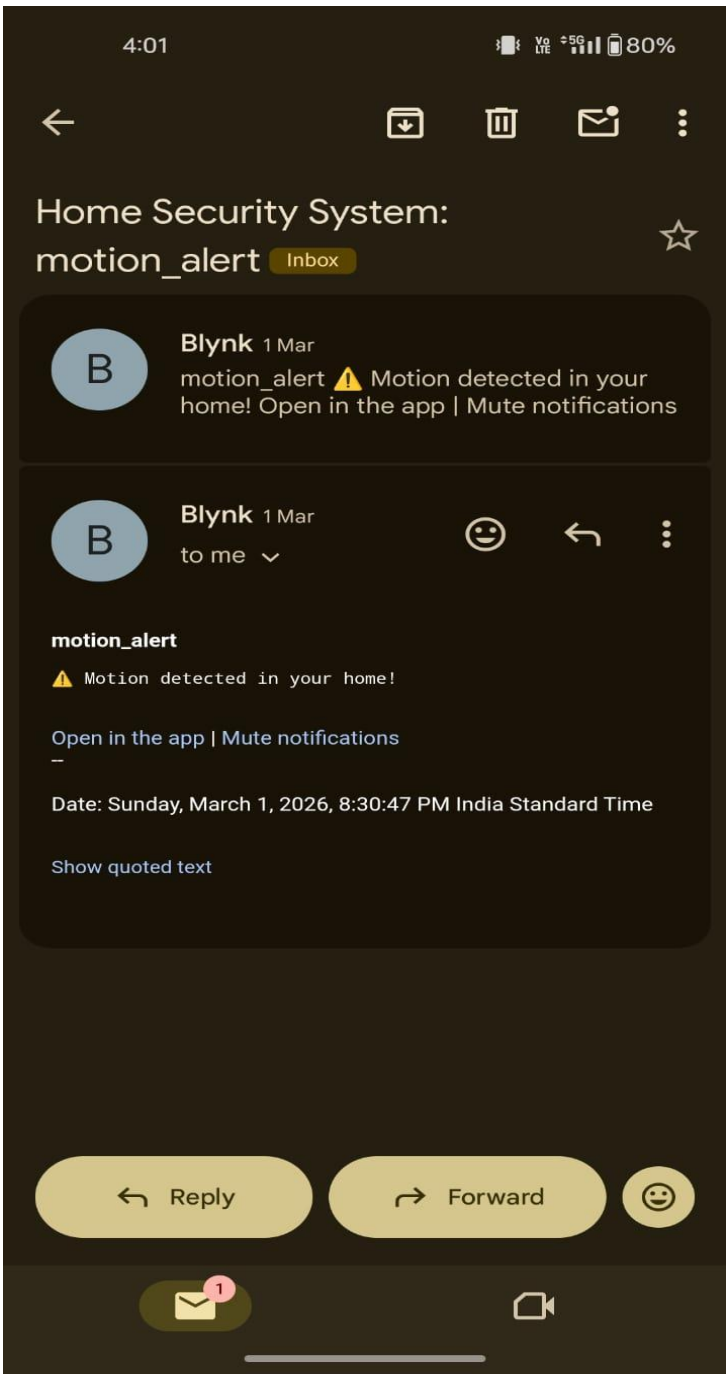
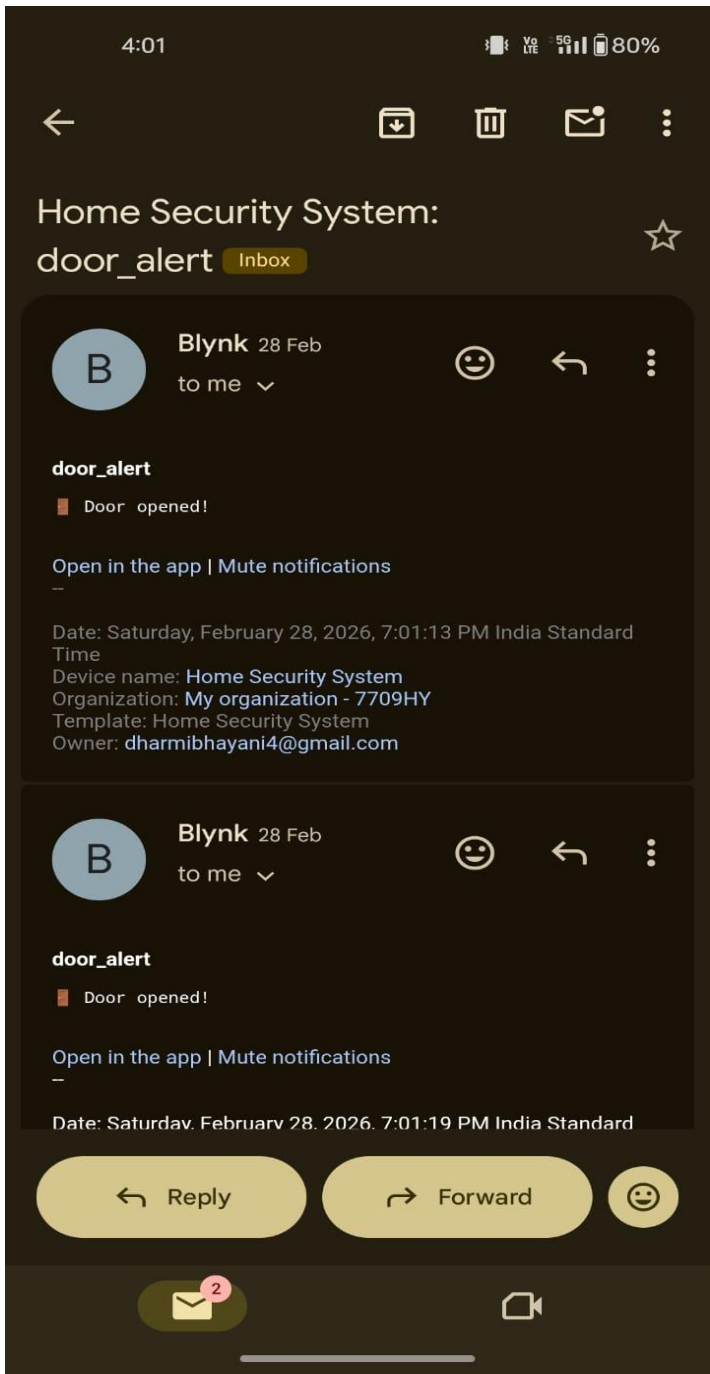
The assessment also examined the responsiveness of the system. When motion or door activity was detected, the NodeMCU processed the signal quickly and activated the buzzer alarm without noticeable delay. This quick response significantly improved user confidence in the system's ability to detect security threats.

Feedback from users indicated that the system was simple, reliable, and effective for basic home security monitoring. Users appreciated the automatic detection feature, which allows the system to operate continuously without manual intervention.

Some suggestions were also provided for improving the system in future versions. These suggestions include adding mobile notifications, smartphone applications, remote monitoring features, and visual indicators such as LEDs to enhance the overall user experience.







### 5.3 Data Security and Integrity

Ensuring data security and maintaining system integrity are important aspects of the IoT-Based Smart Home Security System. Since the system collects and processes information from sensors to detect possible security threats, it is necessary to ensure that the data generated by the system is reliable, accurate, and protected from unauthorized access or manipulation.

The system primarily handles data generated by sensors such as the PIR motion sensor and magnetic door sensor. These sensors send signals to the NodeMCU microcontroller, which processes the information and triggers alerts when suspicious activity is detected. To maintain security, the system ensures that only authorized users can monitor or control the device.

Basic authentication mechanisms can be implemented when connecting the NodeMCU device to a network or IoT platform. This helps prevent unauthorized users from accessing system data or interfering with system operations. Secure network connections also help protect data transmitted between the device and the monitoring platform.

To maintain data integrity, the system ensures that sensor signals are accurately transmitted and processed by the microcontroller. Reliable wiring connections using jumper wires and breadboard connections help prevent signal loss or corruption during data transmission.

Proper programming of the NodeMCU ensures that the received sensor data is processed correctly and that the correct response—such as activating the buzzer alarm—is triggered. The system also maintains event logging, where important security events such as motion detection or door opening can be recorded. These records help users review past security activities and verify the accuracy of system operations.

Additionally, the system is designed to operate in a stable and continuous monitoring mode, ensuring that data generated by sensors remains consistent and reliable over long periods of operation. Regular testing and system checks help detect potential hardware faults or communication errors that could affect data integrity.

## Chapter 6: Conclusion and Future Work

### 6.1 Conclusion

The rapid advancement of Internet of Things (IoT) technology has created new opportunities for developing intelligent systems that improve safety, automation, and convenience in everyday life. The project titled “IoT-Based Smart Home Security System” demonstrates how IoT devices and sensors can be integrated to create an efficient and reliable home security solution.

The main objective of this project was to design and implement a smart security system capable of monitoring the home environment and detecting potential security threats. This was achieved by integrating key hardware components such as the NodeMCU microcontroller, PIR motion sensor, magnetic door sensor, buzzer alarm, breadboard, jumper wires, and USB cable.

These components work together to detect motion or unauthorized door activity and immediately alert the user through an alarm system.

During the development of this project, the system was designed to continuously monitor the surroundings and respond to suspicious activities. The PIR motion sensor detects human movement within a specific range, while the magnetic door sensor detects whether a door or window is opened or closed.

When any abnormal activity is detected, the sensor sends a signal to the NodeMCU microcontroller, which processes the data and activates the buzzer alarm as an alert mechanism.

The project also demonstrated how IoT-based systems can improve traditional home security solutions by providing automatic monitoring and faster response to potential threats. Compared to conventional security systems that rely on manual observation, this system operates automatically and continuously, reducing the risk of delayed responses.

Another important achievement of the project is the successful integration of multiple hardware modules into a single working system. Each component performs a specific role, such as sensing environmental activity, processing sensor data, and triggering alerts. The smooth coordination between sensors and the microcontroller ensures efficient operation and reliable security monitoring.

Testing and evaluation of the system confirmed that it can effectively detect motion and door activity with quick response time and stable performance.

The system operates reliably under normal conditions and provides immediate alerts whenever suspicious activity occurs.

In conclusion, the IoT-Based Smart Home Security System provides a simple, cost-effective, and efficient solution for improving home security. The project demonstrates how IoT technology can be used to create intelligent monitoring systems that enhance safety and convenience. By combining sensors, microcontrollers, and alert mechanisms, the system provides a practical approach to protecting homes from potential intrusions.

The IoT-Based Smart Home Security System developed in this project provides an efficient and reliable solution for monitoring and protecting residential environments. The system integrates various hardware components such as the NodeMCU microcontroller, PIR motion sensor, magnetic door sensor, buzzer, breadboard, jumper wires, and USB cable to detect suspicious activities and alert users immediately.

The main objective of the project was to design a smart security system that can automatically detect motion or unauthorized door access and respond quickly by triggering an alarm. During the development process, the sensors were successfully connected to the NodeMCU microcontroller, which processes the incoming signals and activates the buzzer alarm whenever an intrusion is detected.

## 6.2 Future Work

Although the IoT-Based Smart Home Security System developed in this project performs effectively in detecting motion and door activity, there are several opportunities to enhance and expand the system in the future. With advancements in IoT technology and smart devices, the system can be improved to provide more advanced security features and better user interaction.

One possible improvement is the integration of mobile applications for remote monitoring and control. By connecting the NodeMCU to a smartphone application, users would be able to receive real-time notifications whenever motion or unauthorized door access is detected.

This would allow homeowners to monitor their property from anywhere using the internet. Another potential enhancement is the addition of camera modules to the system. A camera can capture images or video when suspicious activity is detected. These images can be sent directly to the user's mobile device, providing visual confirmation of security events and improving the effectiveness of the system.

The system can also be upgraded by incorporating cloud-based data storage. Storing event logs and security records in cloud servers would allow users to access past security events and analyze system activity over time. This feature would improve data management and help users track potential security threats.

Future versions of the system could also include additional sensors such as smoke detectors, gas sensors, temperature sensors, or vibration sensors. These sensors would allow the system to detect other types of hazards such as fire, gas leakage, or forced entry, thereby expanding the system into a complete smart home safety solution.

Another improvement could involve implementing machine learning or AI-based detection systems that can distinguish between normal activities and suspicious behavior. This would reduce false alarms and improve the accuracy of threat detection.

Furthermore, the system could be enhanced with energy-efficient designs and battery backup systems to ensure continuous operation even during power failures. This would increase system reliability and ensure that security monitoring continues without interruption.

Although the developed system performs its basic security functions effectively, there are several improvements and enhancements that can be implemented in the future to increase its capabilities and efficiency. One possible improvement is the integration of a mobile application that allows users to receive real-time notifications and monitor their home remotely.



This would enable homeowners to check the status of their security system from anywhere using a smartphone.

Another potential enhancement is the addition of camera modules that can capture images or video when motion is detected. This would provide visual evidence of suspicious activity and improve the overall effectiveness of the security system.

The system can also be expanded by incorporating additional sensors such as smoke sensors, gas sensors, or temperature sensors. These sensors would allow the system to detect other hazards like fire, gas leaks, or abnormal environmental conditions, making it a complete smart home safety system.

Future versions of the project could also include cloud storage and data logging features. Storing security events and sensor data in cloud platforms would allow users to review past activities and maintain a record of security alerts.

Another improvement would be the use of artificial intelligence or machine learning algorithms to analyze sensor data and reduce false alarms. Intelligent detection systems could differentiate between normal activities and potential threats more accurately.

In conclusion, while the current system successfully demonstrates the concept of IoT-based home security, future improvements such as mobile integration, camera monitoring, cloud storage, additional sensors, and intelligent detection systems can significantly enhance its functionality and reliability. These advancements would transform the system into a more advanced smart home automation and security platform.

## 6.3 Limitations

Although the IoT-Based Smart Home Security System successfully achieved its main objective of detecting motion and door activity, some limitations were observed during the development and testing stages. These limitations may affect the overall performance of the system and can be improved in future versions.

### 1. Hardware Dependency

The system relies heavily on hardware components such as the NodeMCU microcontroller, PIR motion sensor, magnetic door sensor, and buzzer. If any of these components fail or malfunction, the system may not operate correctly. Hardware damage, loose connections, or power supply issues can interrupt the system's operation.

### 2. Limited Detection Range

The PIR motion sensor has a limited detection range and angle. If movement occurs outside the sensor's coverage area, the system may fail to detect it. Similarly, the magnetic door sensor only detects door opening and closing but cannot detect other types of intrusion.

### 3. False Alarms

The motion sensor may sometimes detect movement caused by pets, insects, or environmental changes such as temperature variations. This can result in false alarms, which may reduce the reliability of the system.

### 4. Dependence on Power Supply

The system requires a continuous power supply through the USB cable or external power source. In case of power failure, the system will stop functioning unless a backup power source is available.

### 5. Limited Remote Monitoring

In its current form, the system mainly triggers a local buzzer alarm when suspicious activity is detected. Without additional IoT platforms or mobile applications, the system does not provide advanced remote monitoring features.

### 6. Limited Scalability

The current design is suitable for small home security setups, but expanding it to cover larger areas or multiple rooms would require additional sensors, controllers, and improved network communication.

### 7. Environmental Factors

The performance of sensors may be affected by environmental conditions such as high temperature, dust, humidity, or poor sensor placement. These factors can influence the accuracy and reliability of motion detection.

#### 8. Lack of Visual Monitoring

The present system does not include a camera module, which means it cannot capture images or video of the detected event. Therefore, users cannot visually confirm the cause of the alarm.

#### 9. Network Connectivity Issues

If the system is connected to an IoT platform or Wi-Fi network, unstable internet connectivity may affect communication between the NodeMCU microcontroller and the monitoring system. Poor network connectivity can delay notifications or prevent remote monitoring.

#### 10. Sensor Placement Sensitivity

The effectiveness of the system depends greatly on the proper placement of sensors. If the PIR motion sensor or magnetic door sensor is installed in an incorrect position, the system may fail to detect motion or door activity accurately.

#### 11. Environmental Interference

External environmental factors such as temperature changes, dust, humidity, or direct sunlight may affect the performance of the sensors. These factors can sometimes reduce detection accuracy.

#### 12. Limited Security Features

The current system mainly detects motion and door activity. It does not include advanced security features such as facial recognition, camera monitoring, or biometric authentication, which are available in more advanced smart security systems.

## Chapter 7: References

1. “Internet of Things (IoT) in Smart Home Security Systems” – International Journal of Computer Applications (IJCA)
  - Authors: Various researchers in IoT and embedded systems
  - Focus: Discusses the implementation of IoT technology in home automation and security monitoring using sensors and microcontrollers.
2. “NodeMCU ESP8266 for IoT Based Applications” – IEEE Research Paper
  - Authors: Embedded systems researchers
  - Focus: Explains how the NodeMCU microcontroller can be used for IoT-based projects including automation and security systems.
3. “Design and Implementation of Smart Home Security System Using IoT” – International Journal of Engineering Research & Technology (IJERT)
  - Authors: Various IoT researchers
  - Focus: Describes the design of a smart home security system using motion sensors, microcontrollers, and alert mechanisms.
4. “PIR Motion Sensor Based Security System” – International Journal of Scientific & Engineering Research (IJSER)
  - Authors: Electronics and embedded system researchers
  - Focus: Explains the working principle of PIR motion sensors and their use in security systems for detecting human movement.
  - Good because: Provides technical understanding of motion detection technology.
5. “Smart Home Automation Using IoT” – Research Article from IEEE Xplore
  - Authors: IoT and automation researchers
  - Focus: Discusses IoT applications in home automation, including monitoring, security, and remote control systems.
6. “Magnetic Door Sensor Based Security System” – Electronics Research Study
  - Authors: Embedded system developers
  - Focus: Describes how magnetic sensors are used to detect door opening and closing in security systems.
7. “Internet of Things: A Survey on Applications and Future Trends” – IEEE Communications Surveys
  - Authors: IoT researchers and technology experts
  - Focus: Provides an overview of IoT applications including smart homes, smart cities, healthcare, and security systems.